

Combinational Circuits

Digital Logic Design

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Introduction

❖ Combinational

- The outputs depend only on the current input values
- It uses only logic gates

❖ Sequential

- ❑ The outputs depend on the current and past input values
- ❑ It uses logic gates and storage elements
- ❑ Example
 - Vending machine
- ❑ They are referred as finite state machines since they have a finite number of states

Block diagram of Sequential Circuit

- ❖ Memory elements can store binary information
 - This information at any given time determines the state of the circuit at that time

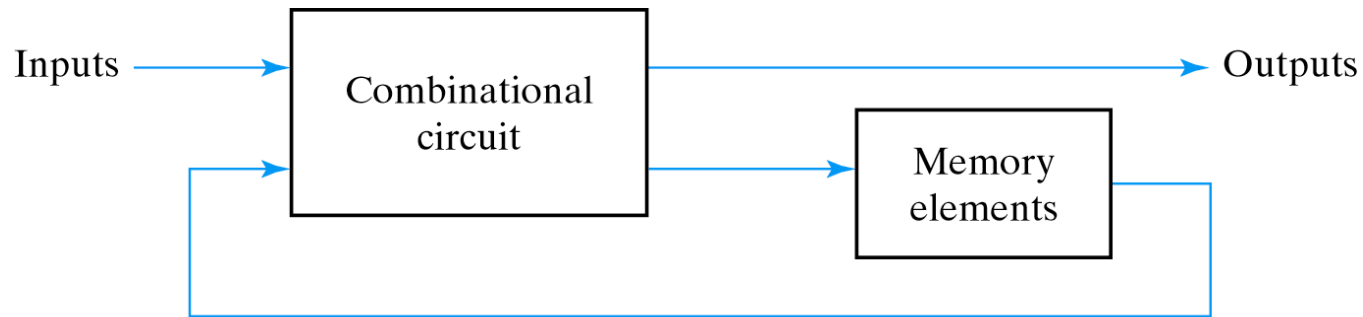


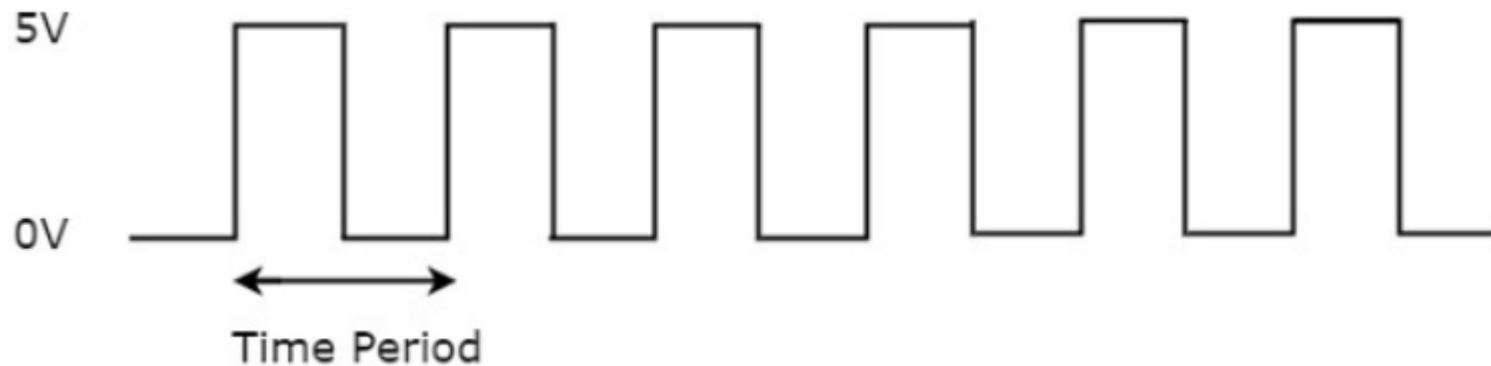
Fig. 5-1 Block Diagram of Sequential Circuit

Difference between Flip-flop and Latch

Flip-Flop	Latch
Flip-flop is a bistable device i.e., it has two stable states that are represented as 0 and 1.	Latch is also a bistable device whose states are also represented as 0 and 1.
It checks the inputs but changes the output only at times defined by the clock signal or any other control signal.	It checks the inputs continuously and responds to the changes in inputs immediately.
It is a edge triggered device.	It is a level triggered device.
Gates like NOR , NOT , AND , NAND are building blocks of flip flops.	These are also made up of gates.
They are classified into asynchronous or synchronous flipflops.	There is no such classification in latches
It forms the building blocks of many sequential circuits like counters .	These can be used for the designing of sequential circuits but are not generally preferred.
Flip-flop always have a clock signal	Latches doesn't have a clock signal
Flip-flop can be build from Latches	Latches can be build from gates
ex: D Flip-flop , JK Flip-flop	ex:SR Latch, D Latch

Clock Signals

- It emits a series of pulses with a precise pulse width and precise interval between consecutive pulses
- Timing interval between the corresponding edges of two consecutive pulses is known as the clock cycle time, or period



Clock Signals – Level triggered

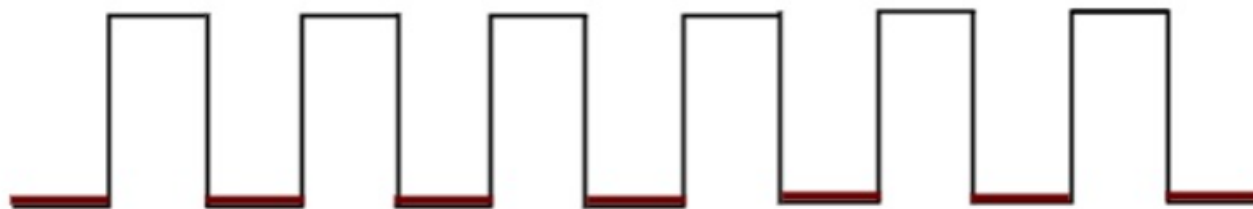
- There are two levels, namely logic High and logic Low in clock signal. Following are the two **types of level triggering**.
- Positive Level Triggering
- Negative Level Triggering

Clock Signals – Level triggered

If the sequential circuit is operated with the clock signal when it is in **Logic High**, then that type of triggering is known as **Positive level triggering**. It is highlighted in below figure.



If the sequential circuit is operated with the clock signal when it is in **Logic Low**, then that type of triggering is known as **Negative level triggering**. It is highlighted in the following figure.



Clock Signals – Edge triggered

- There are two types of transitions that occur in clock signal. That means, the clock signal transitions either from Logic Low to Logic High or Logic High to Logic Low.
- Following are the two **types of edge triggering** based on the transitions of clock signal.
- Positive Edge Triggering
- Negative Edge Triggering

Clock Signals – Edge triggered

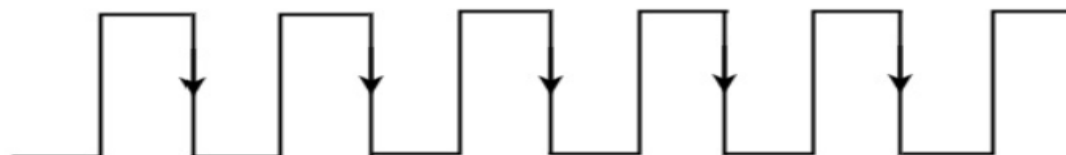
Positive Edge Triggering

If the sequential circuit is operated with the clock signal that is transitioning from Logic Low to Logic High, then that type of triggering is known as **Positive Edge Triggering**. It is also called as rising edge triggering. It is shown in the following figure.

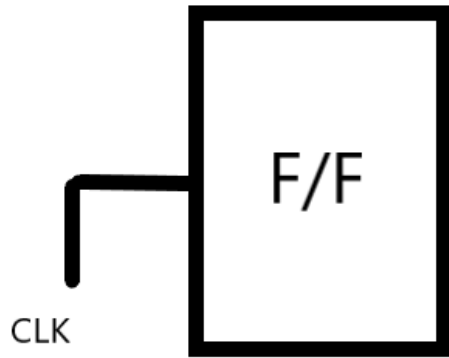


Negative Edge Triggering

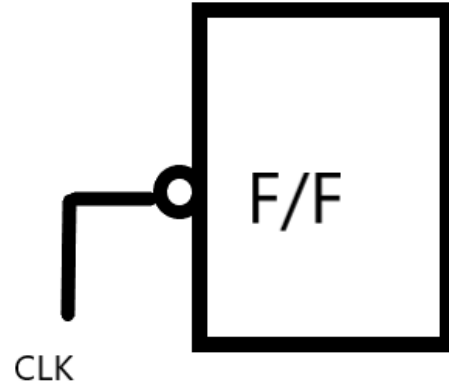
If the sequential circuit is operated with the clock signal that is transitioning from Logic High to Logic Low, then that type of triggering is known as **Negative Edge Triggering**. It is also called as falling edge triggering. It is shown in the following figure.



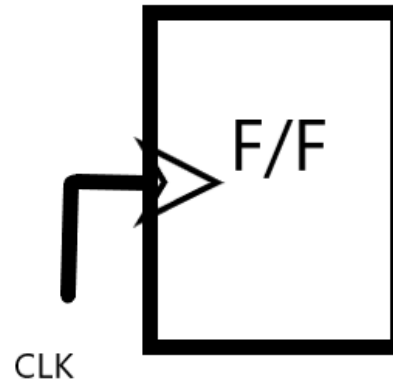
Clock Signals



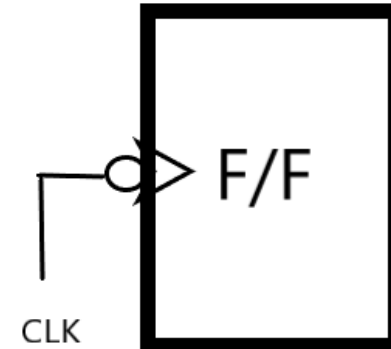
Positive level
trigger



Negative
level
trigger



Positive
Edge
trigger

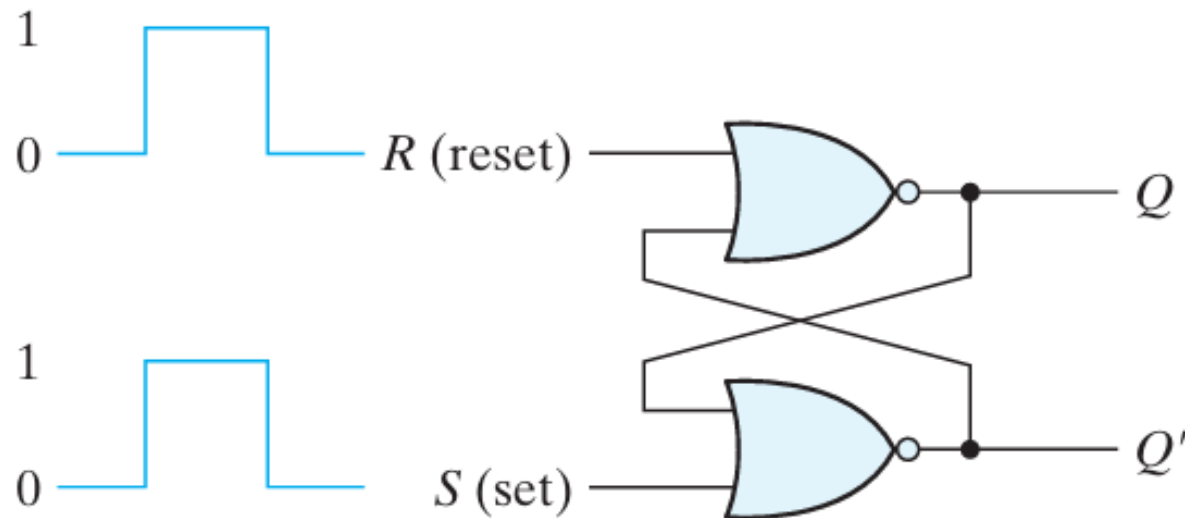


Negative
Edge
trigger

SR Latch

- S-R latches i.e., Set-Reset latches are the simplest form of latches and are implemented using two inputs: S (Set) and R (Reset).
- The S input sets the output to 1, while the R input resets the output to 0.
- When both S and R inputs are at 1, the latch is said to be in an "undefined" state.
- They are also known as preset and clear states. The SR latch forms the basic building blocks of all other types of flip-flops.

SR Latch-NOR Gate



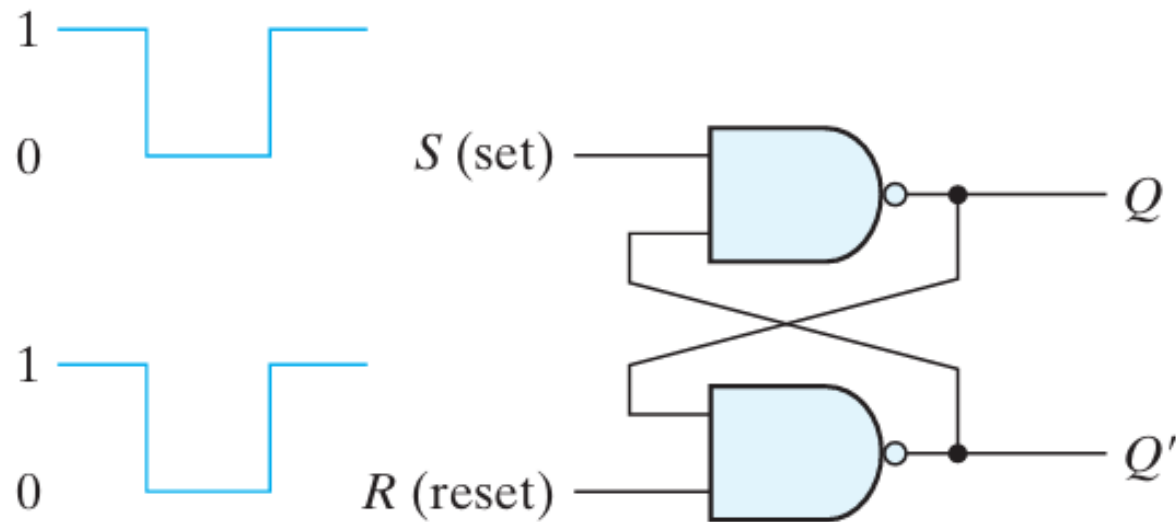
(a) Logic diagram

S	R	Q	Q'	
1	0	1	0	
0	0	1	0	(after $S = 1, R = 0$)
0	1	0	1	
0	0	0	1	(after $S = 0, R = 1$)
1	1	0	0	(forbidden)

(b) Function table

FIGURE 5.3
 SR latch with NOR gates

SR Latch-Nand Gate



(a) Logic diagram

S	R	Q	Q'
1	0	0	1
1	1	0	1 (after $S = 1, R = 0$)
0	1	1	0
1	1	1	0 (after $S = 0, R = 1$)
0	0	1	1 (forbidden)

(b) Function table

FIGURE 5.4

SR latch with NAND gates

SR Flip Flop

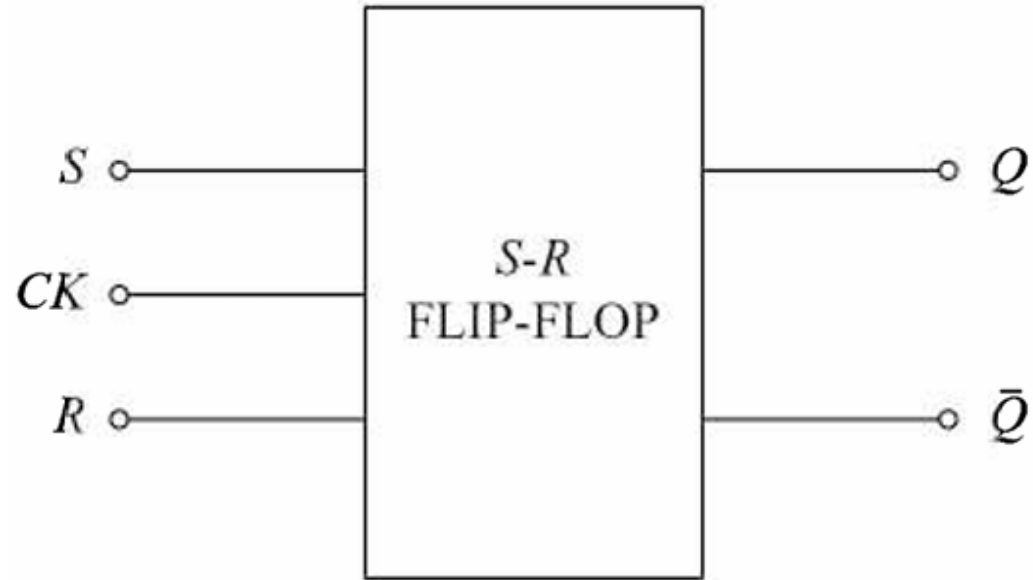


Fig. 7.6 **Logic Symbol of Clocked S-R**

FLIP-FLOP

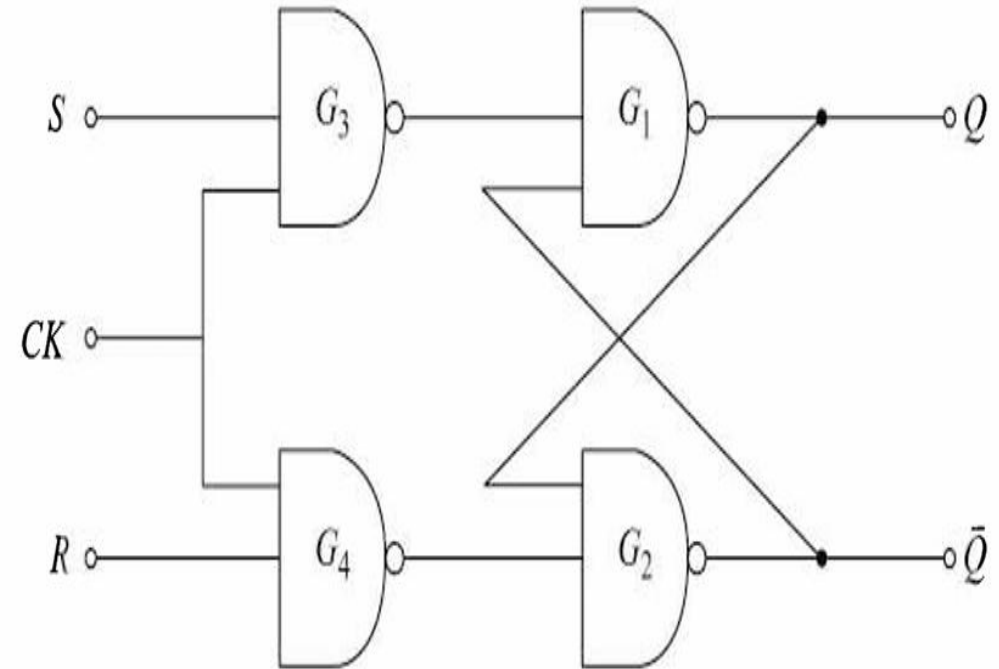


Fig. 7.5 **A Clocked S-R FLIP-FLOP**
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SR Flip Flop

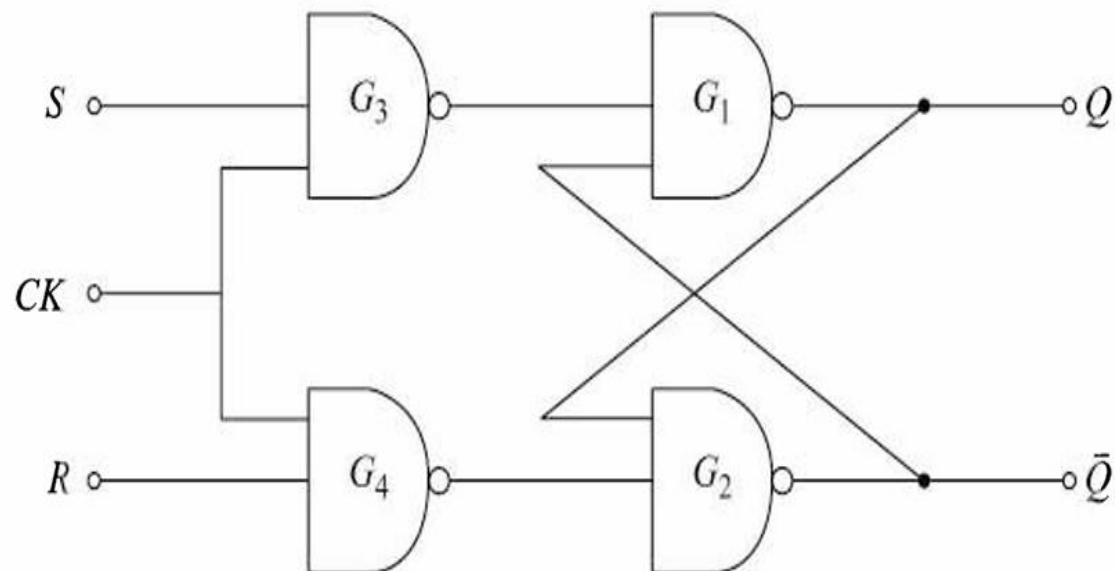
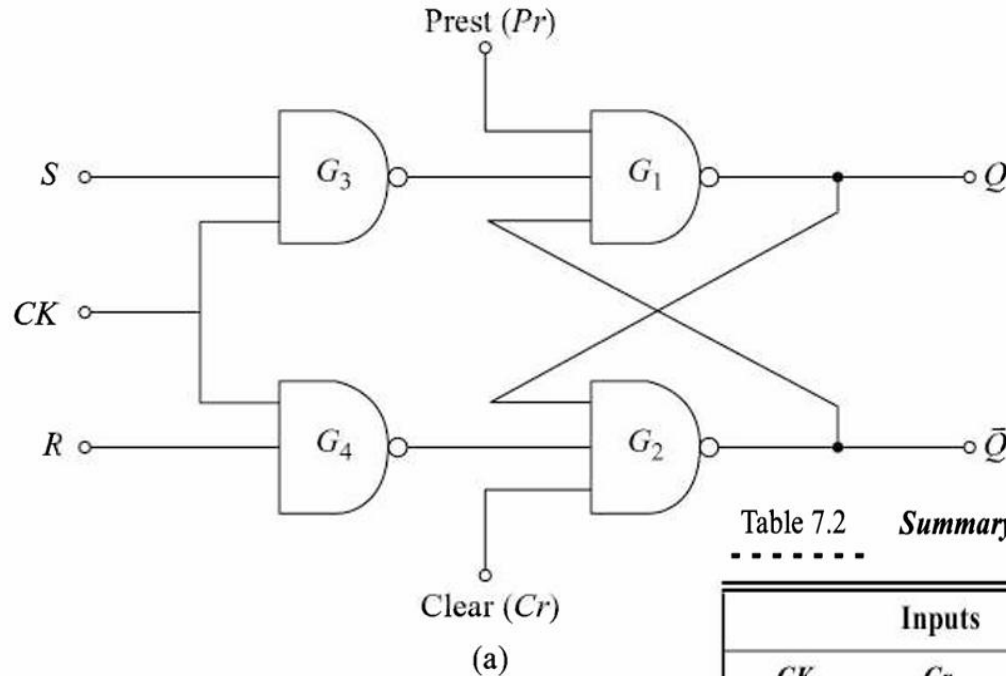


Fig. 7.5 *A Clocked S-R FLIP-FLOP*

Table 7.1 *Truth Table of S-R FLIP-FLOP*

Inputs		Output
S_n	R_n	Q_{n+1}
0	0	Q_n
1	0	1
0	1	0
1	1	?

SR Flip Flop



Preset is used to Set the circuit '1'.
Clear is used to reset the circuit

Table 7.2 Summary of Operation of S-R FLIP-FLOP

Inputs			Output Q	Operation performed
CK	Cr	Pr		
1	1	1	Q_{n+1} (Table 7.1)	Normal FLIP-FLOP
0	0	1	0	Clear
0	1	0	1	Preset

SR Flip Flop

Characteristic Table

S	R	Q _n	Q(n+1)
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	X
1	1	1	X

S	RQ _n	R'Q _n			
		00	01	11	10
s' 0			1		
		0	1	3	2
s 1		1	1	X	X
		4	5	7	6

SR Flip Flop K-Map

$$Q(n+1) = S + R'Q_n$$

SR Flip Flop

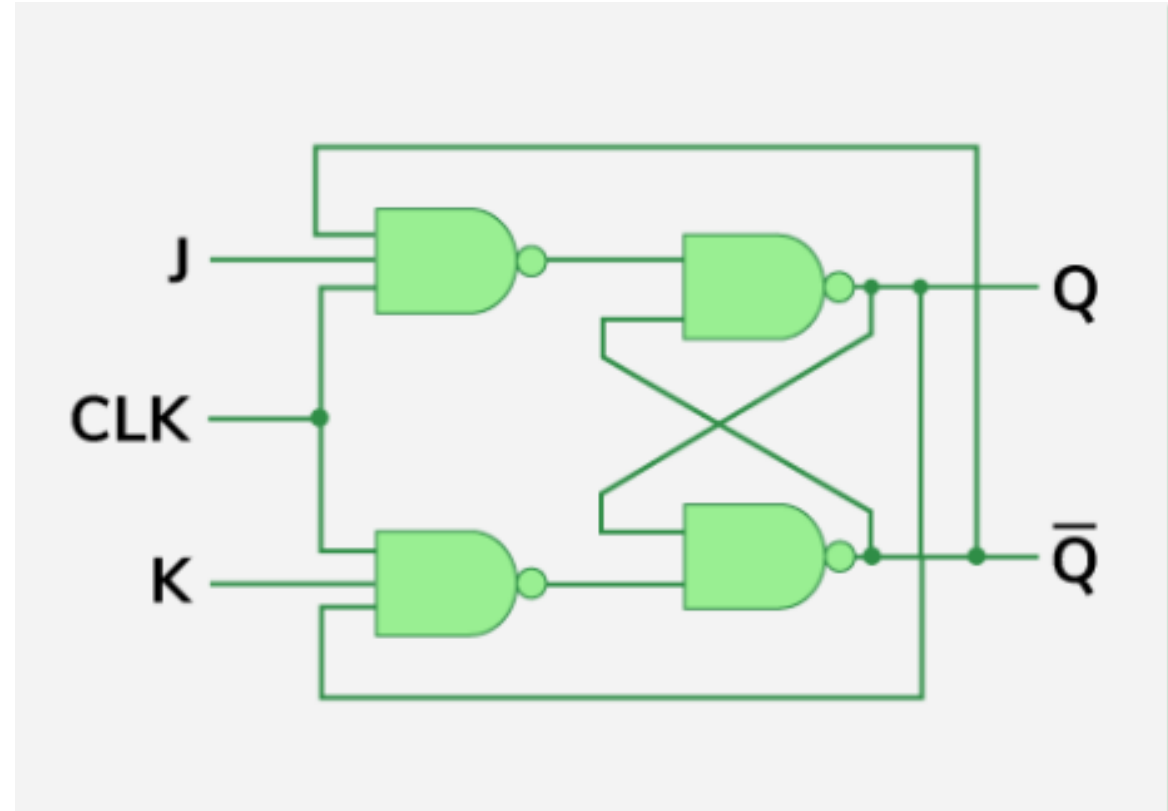
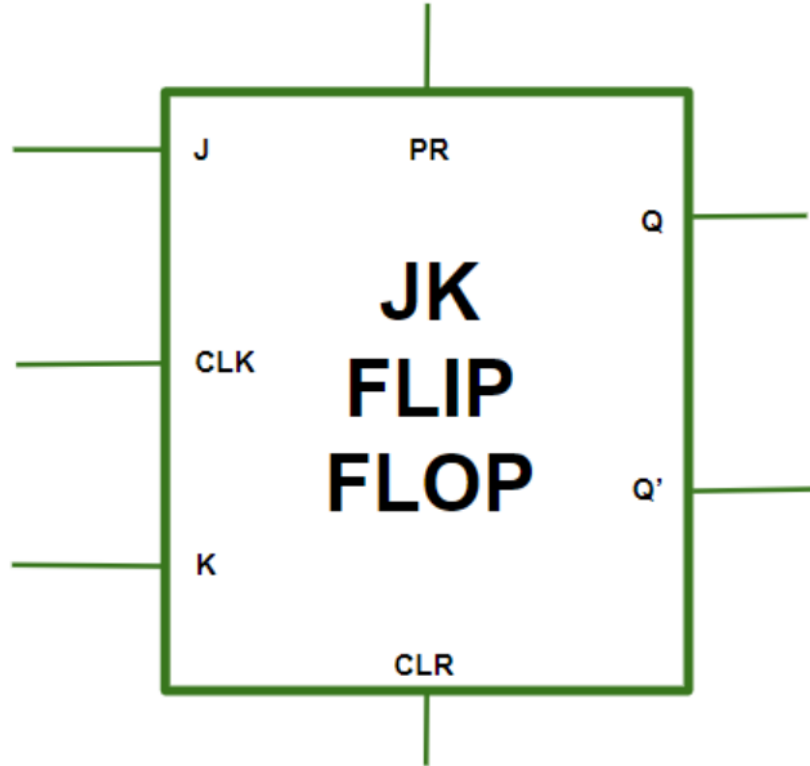
- Excitation Table basically tells about the excitation which is required by flip flop to go from current state to next state.

Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

JK Flip-Flop

- The uncertainty in the state of an S–R FLIP-FLOP when $S_n = R_n = 1$ can be eliminated by converting it into a J–K FLIP-FLOP.
- The data inputs are J and K which are ANDed with $\bar{Q}$ and Q respectively, to obtain S and R inputs, i.e.
- $S = J \cdot \bar{Q}$
- $R = K \cdot Q$

JK Flip-Flop



JK Flip-Flop

Table 7.3b *Truth Table of J-K FLIP-FLOP*

Inputs		Output
J_n	K_n	Q_{n+1}
0	0	Q_n
1	0	1
0	1	0
1	1	$\bar{Q}_n$

JK Flip-Flop

INPUT			OUTPUT
J	K	Q _n	Q _(n+1)
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

JK Flip-Flop

Characteristic Equation: $\bar{K}Q_n + J\bar{Q}_n$

$J \backslash KQ_n$					
		$\bar{K}\bar{Q}_n$	$\bar{K}Q_n$	KQ_n	$K\bar{Q}_n$
$\bar{J}$	0	1	0	0	0
J	1	1	0	1	

$J \backslash KQ_n$					
		00	01	11	10
0	0	1	0	0	0
1	1	1	0	1	

JK Flip-Flop

Group 1: $\bar{J}\bar{K}Qn + J\bar{K}Qn$

Group 2: $J\bar{K}\bar{Q}n + JK\bar{Q}n$

Equation: $\bar{J}\bar{K}Qn + J\bar{K}Qn + J\bar{K}\bar{Q}n + JK\bar{Q}n$

$= (\bar{J} + J)\bar{K}Qn + J\bar{Q}n(\bar{K} + K) \dots (\text{Distribution Law})$

$= 1.\bar{K}Qn + 1.J\bar{Q}n \dots (\text{Compliment Law, } \bar{J} + J = 1 \text{ and } \bar{K} + K = 1)$

$= \bar{K}Qn + J\bar{Q}n \dots (\text{Identity Law, } 1 \& A = A)$

JK Flip-Flop

INPUT		OUTPUT	
Q _n	Q(n+1)	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Race Around Condition

- The difficulty of both inputs 1 ($S = R = 1$) being not allowed in an S–R FLIP-FLOP is eliminated in a J–K FLIP-FLOP by using the feedback connection from outputs to the inputs of the gates
- Table assumes that the inputs do not change during the clock pulse ($CK = 1$), which is not true because of the feedback connections.
- Consider, for example, that the inputs are $J = K = 1$ and $Q = 0$, and a pulse as shown in Fig. 7.11 is applied at the clock input.
- After a time interval Δt equal to the propagation delay through two NAND gates in series, the output will change to $Q = 1$.
- Now we have $J = K = 1$ and $Q = 1$ and after another time interval of Δt the output will change back to $Q = 0$.
- Hence, we conclude that for the duration t_p of the clock pulse, the output will oscillate back and forth between 0 and 1.
- At the end of the clock pulse, the value of Q is uncertain. This situation is referred to as the race-around condition.

Race Around Condition

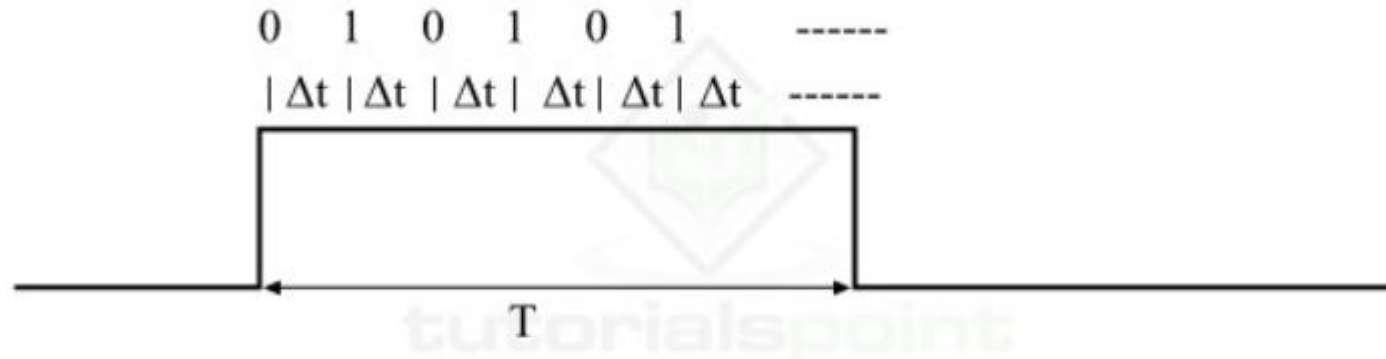


Figure 2 - Race Around Condition in JK Flip Flop

- The race-around condition can be avoided if $t_p < \Delta t < T$.
- However, it may be difficult to satisfy this inequality because of very small propagation delays in ICs.
- A more practical method for overcoming this difficulty is the use of the master-slave (M-S) configuration discussed below.

Master Slave JK Flip Flop

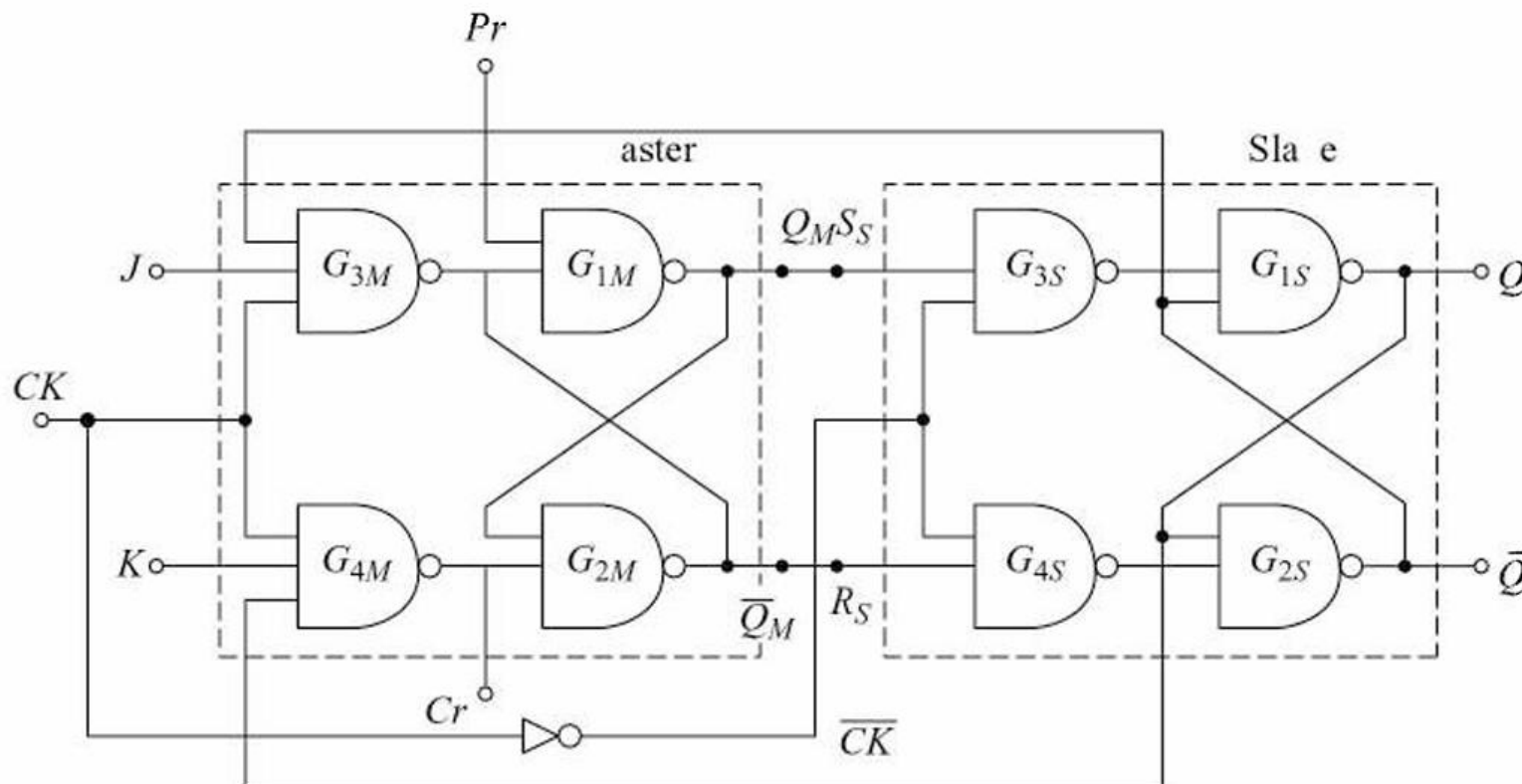


Fig. 7.12 A Master-Slave J-K FLIP-FLOP

Master Slave JK Flip Flop

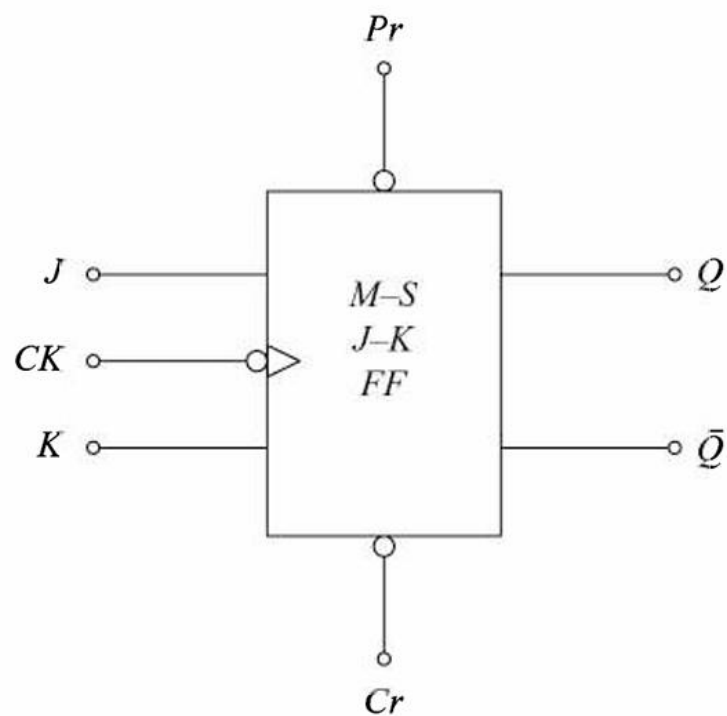


Fig. 7.13

*A Master-Slave J-K FLIP-FLOP
Logic Symbol*

Master Slave JK Flip Flop

- The Master-Slave JK Flip-Flop is made of two JK flip-flops in series.
- One flip-flop acts as the master, the other as the slave.
- Master output $\rightarrow$ Slave inputs, and slave output $\rightarrow$ Master inputs.
- An inverter provides the inverted clock pulse to the slave.
- If $CP = 1$ for master, $CP = 0$ for slave.
- If $CP = 0$ for master, $CP = 1$ for slave.

D Flip Flop

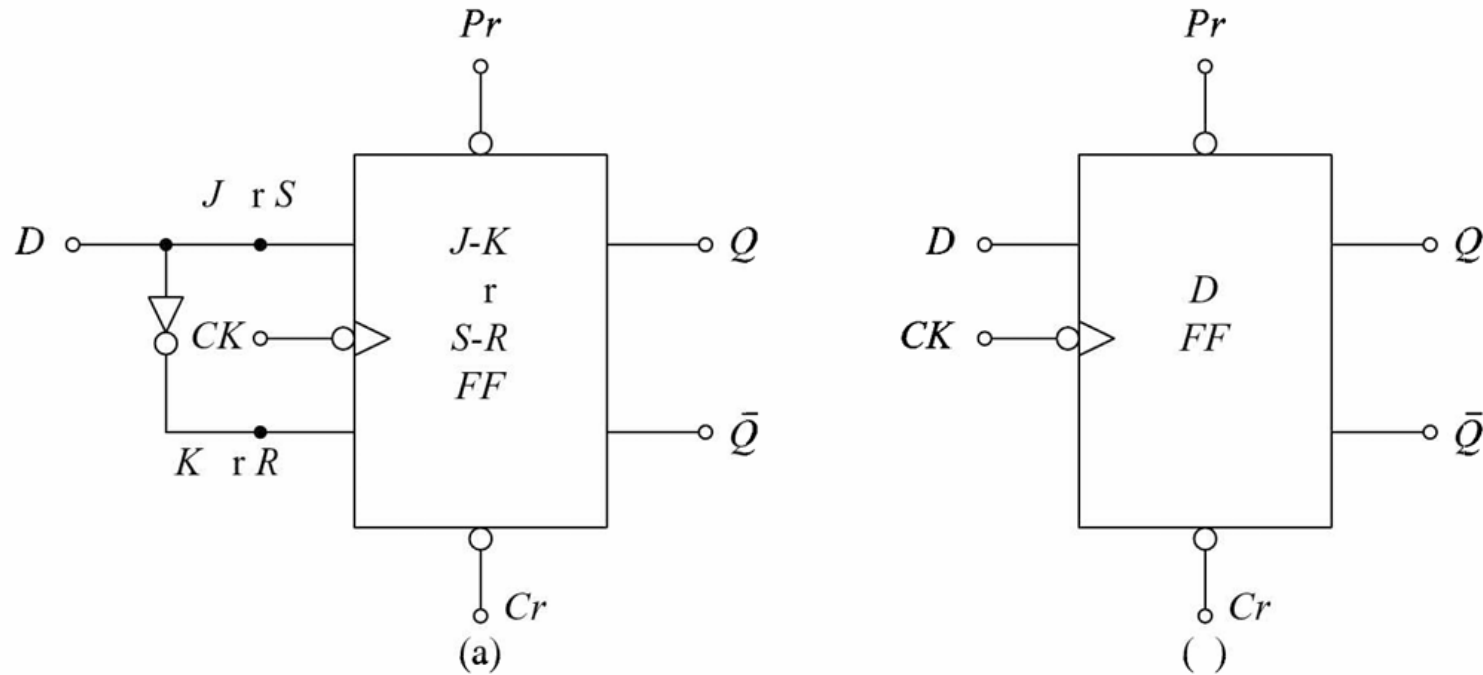


Fig. 7.14 (a) A J-K or S-R FLIP-FLOP Converted into a D-type FLIP-FLOP (b) its Logic Symbol

D Flip Flop

- D flip flop is an electronic devices that is known as "**delay flip flop**" or "**data flip flop**" which is used to store single bit of data.
- D flip flops can be either synchronous or asynchronous.
- The clock signal is required for the synchronous version of D flip flops but not for the asynchronous one.
- The D flip flop has two inputs, data and clock input which controls the flip flop.
- When **clock input is high**, the data is transferred to the **output** of the flip flop.
- When the **clock input is low**, the output of the flip flop is held in its **previous state**.

D Flip Flop

Table 7.4 *Truth Table of a D-type* FLIP-FLOP

Input D_n	Output Q_{n+1}
0	0
1	1

D Flip Flop

D	Q(Current)	Q(n+1) (Next)
0	0	0
0	1	0
1	0	1
1	1	1

Characteristics table of D Flip Flop

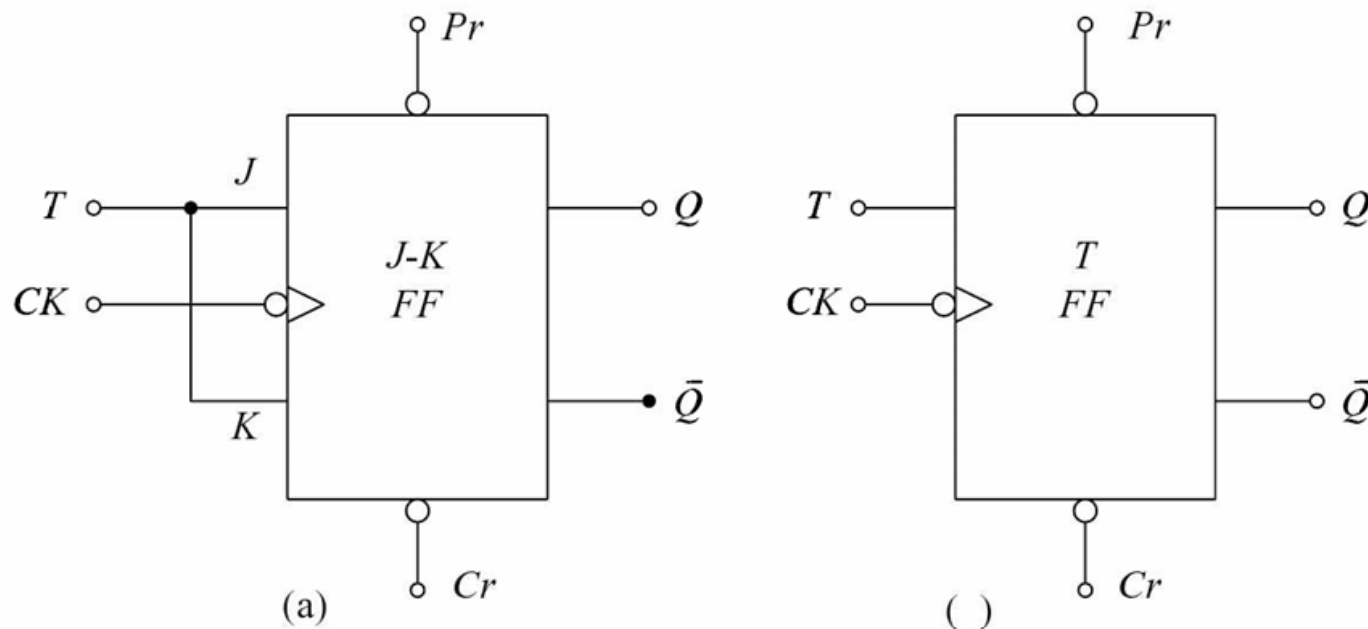
$$Q(n+1) = D(n)$$

D Flip Flop

Excitation Table

Present State	Next State	<i>D-FF</i> D_n
0	0	0
0	1	1
1	0	0
1	1	1

T-Flip Flop



g. 7.15

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(a) A J-K FLIP-FLOP *Converted into a T-type* FLIP-FLOP (b) its *Logic Symbol*

T-Flip Flop

- **T flip flop** is known as **Toggle Flip Flop** because it is able to toggle its output depending upon on the input.
- Toggle basically indicates that the bit will be flipped i.e., either from 1 to 0 or from 0 to 1.
- Here, a clock pulse is supplied to operate this flip flop, hence it is a clocked flip-flop

Table 7.5 *Truth Table of T-type* FLIP-FLOP

Input T_n	Output Q_{n+1}
0	Q_n
1	$\bar{Q}_n$

T-Flip Flop

T	Q	Q(t+1)
0	0	0
1	0	1
0	1	1
1	1	0

Truth Table of T Flip Flop

	Q_n	Q_n'	Q_n
T	0	1	
T'	0		1
T	1	1	

K-Map for T Flip Flop

$$Q(n+1) = TQ_n' + T'Q_n = T \oplus Q_n$$

T-Flip Flop

Q_t	$Q(t+1)$	T
0	0	0
1	0	1
0	1	1
1	1	0

Excitation Table for T Flip Flop

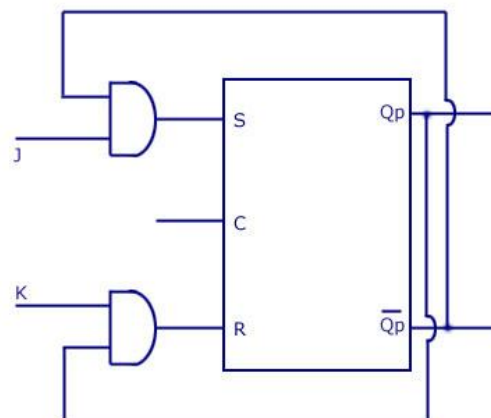
Conversion of the Flip Flop

S-R Flip Flop to J-K Flip Flop

Conversion Table

J-K Inputs		Outputs		S-R Inputs	
J	K	Q _p	Q _{p+1}	S	R
0	0	0	0	0	X
0	0	1	1	X	0
0	1	0	0	0	X
0	1	1	0	0	1
1	0	0	1	1	0
1	0	1	1	X	0
1	1	0	1	1	0
1	1	1	0	0	1

Logic Diagram



J \ KQ _p	00	01	11	10
0	0 ⁰	X ¹	0 ³	0 ²
1	1 ⁴	X ⁵	0 ⁷	1 ⁶

$$S = \bar{J}Q_p$$

J \ KQ _p	00	01	11	10
0	X ⁰	0 ¹	1 ³	X ²
1	0 ⁴	0 ⁵	1 ⁷	0 ⁶

$$R = KQ_p$$

K-Map

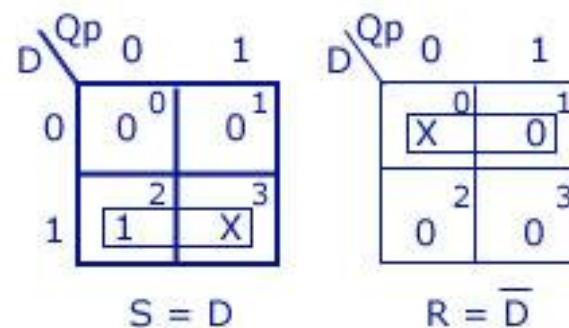
Conversion of the Flip Flop

S-R Flip Flop to D Flip Flop

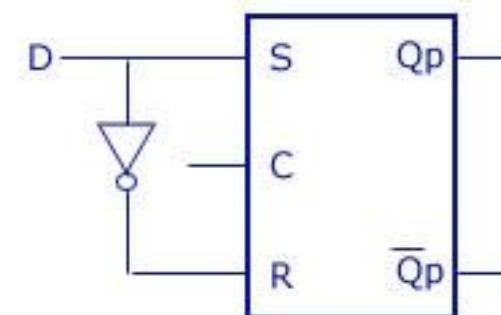
Conversion Table

D Input	Outputs		S-R Inputs	
	Q _p	Q _{p+1}	S	R
0	0	0	0	X
0	1	0	0	1
1	0	1	1	0
1	1	1	X	0

K-maps



Logic Diagram



Counters

- In digital electronics, a set of flip-flops that changes its states in response to pulses applied at the input is called a digital counter.
- In the digital counter circuit, the flip flops are connected in such a way that their combined state at any time is the binary equivalent of the total number of pulses that have occurred up to that time.
- Therefore, as its name implies, a counter is used to count pulses in a digital system.

Counters

Digital counters are classified into the following two types namely,

1. Asynchronous Counter – The type of counter in which the flip flops are connected in such a way that they are not triggered simultaneously is known as asynchronous counter. Asynchronous counter is also known as ripple counter.
2. Synchronous Counter – The type of counter in which all the flip flops are clocked simultaneously is known as a synchronous counter.

Counters

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2. Synchronous Counter – The type of counter in which all the flip flops are clocked simultaneously is known as a synchronous counter.

Counters

There are two basic types of counter

1. Up-Counter (CTU)
2. Down-Counter (CTD)
 - In Up counter when contact change over takes place accumulator value increments by one. Counting is from zero to pre-determined value.
 $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \dots$
 - While in down counter when changeover takes place accumulator value decrements by one. The count is from pre-determined value to zero.
 $7 \rightarrow 6 \rightarrow 5 \rightarrow 4 \dots$

Counters

Design a 3 bit synchronous up counter using JK Flip-Flop

Video Link for reference:

<https://youtu.be/10DVImrPRvM?si=Fps6pWeuUnVftpkp>

Design a 3 bit synchronous up counter using JK Flip-Flop

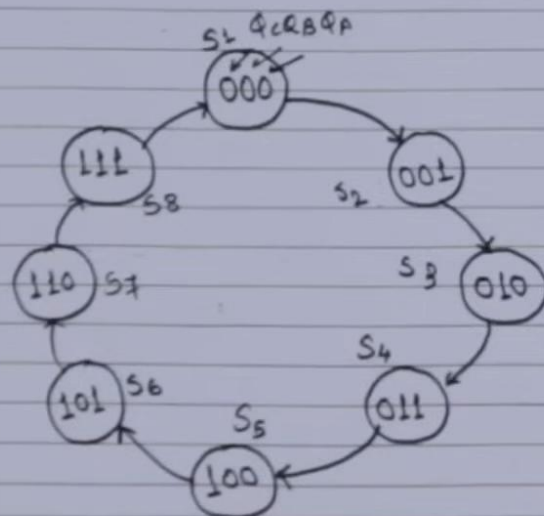
3 bit synchronous up counter using JK-FF

- 1) Number of flip flop used = 3 FF = JK-FF
- 2) Excitation table for JK-FF.
- 3) State diagram = $2^n = 2^3 = 8$
- 4) Truth Table
- 5) K-map
- 6) Logical diagram

Excitation table for JK-FF

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

State diagram.



Truth Table

Q_C	Q_B	Q_A	Q_C^*	Q_B^*	Q_A^*	J_C	K_C	J_B	K_B	J_A	K_A
0	0	0	0	0	1	0	X	0	X	1	X
0	0	1	0	1	0	0	X	1	X	X	1
0	1	0	0	1	1	0	X	X	0	1	X
0	1	1	1	0	0	1	X	X	1	X	1
1	0	0	1	0	1	X	0	0	X	1	X
1	0	1	1	1	0	X	0	1	X	X	1
1	1	0	1	1	1	X	0	X	0	1	X
1	1	1	0	0	0	X	1	X	1	X	1

Design a 3 bit synchronous up counter using JK Flip-Flop

Truth Table

J_C	K_C	J_B	K_B	J_A	K_A
0	X	0	X	1	X
0	X	1	X	X	1
0	X	X	0	1	X
1	X	X	1	X	1
X	0	0	X	1	X
X	0	1	X	X	1
X	0	X	0	1	X
X	1	X	1	X	1

K-map for J_C

$Q_C Q_B$	00	01	11	10
Q_A	0	0	X	X
	1	1	X	X

$$J_C = Q_B Q_A$$

K-map for K_C

$Q_C Q_B$	00	01	11	10
Q_A	X	X	1	1
	1	X	X	1

$$K_C = Q_A Q_B$$

K-map for J_B

$Q_C Q_B$	00	01	11	10
Q_A	0	X	X	1
	1	X	X	1

$$J_B = Q_A$$

K-map for J_A

$Q_C Q_B$	00	01	11	10
Q_A	1	1	1	1
	1	X	X	X

$$J_A = 1$$

K-map for K_A

$Q_C Q_B$	00	01	11	10
Q_A	X	X	X	X
	1	1	1	1

$$K_A = 1$$

K-map for K_B

$Q_C Q_B$	00	01	11	10
Q_A	X	1	1	X
	1	X	1	X

$$\therefore K_B = Q_A$$

Design a 3 bit synchronous up counter using JK Flip-Flop

Truth Table

J_C	K_C	J_B	K_B	J_A	K_A
0	X	0	X	1	X
0	X	1	X	X	1
0	X	X	0	1	X
1	X	X	1	X	1
X	0	0	X	1	X
X	0	1	X	X	1
X	0	X	0	1	X
X	1	X	1	X	1

K-map for J_C

$Q_C Q_B$	00	01	11	10
Q_A	0	0	X	X
	1	1	X	X

$$J_C = Q_B Q_A$$

K-map for K_C

$Q_C Q_B$	00	01	11	10
Q_A	X	X	1	1
	1	X	X	1

$$K_C = Q_A Q_B$$

K-map for J_B

$Q_C Q_B$	00	01	11	10
Q_A	0	X	X	1
	1	X	X	1

$$J_B = Q_A$$

K-map for J_A

$Q_C Q_B$	00	01	11	10
Q_A	1	1	1	1
	1	X	X	X

$$J_A = 1$$

K-map for K_A

$Q_C Q_B$	00	01	11	10
Q_A	X	X	X	X
	1	1	1	1

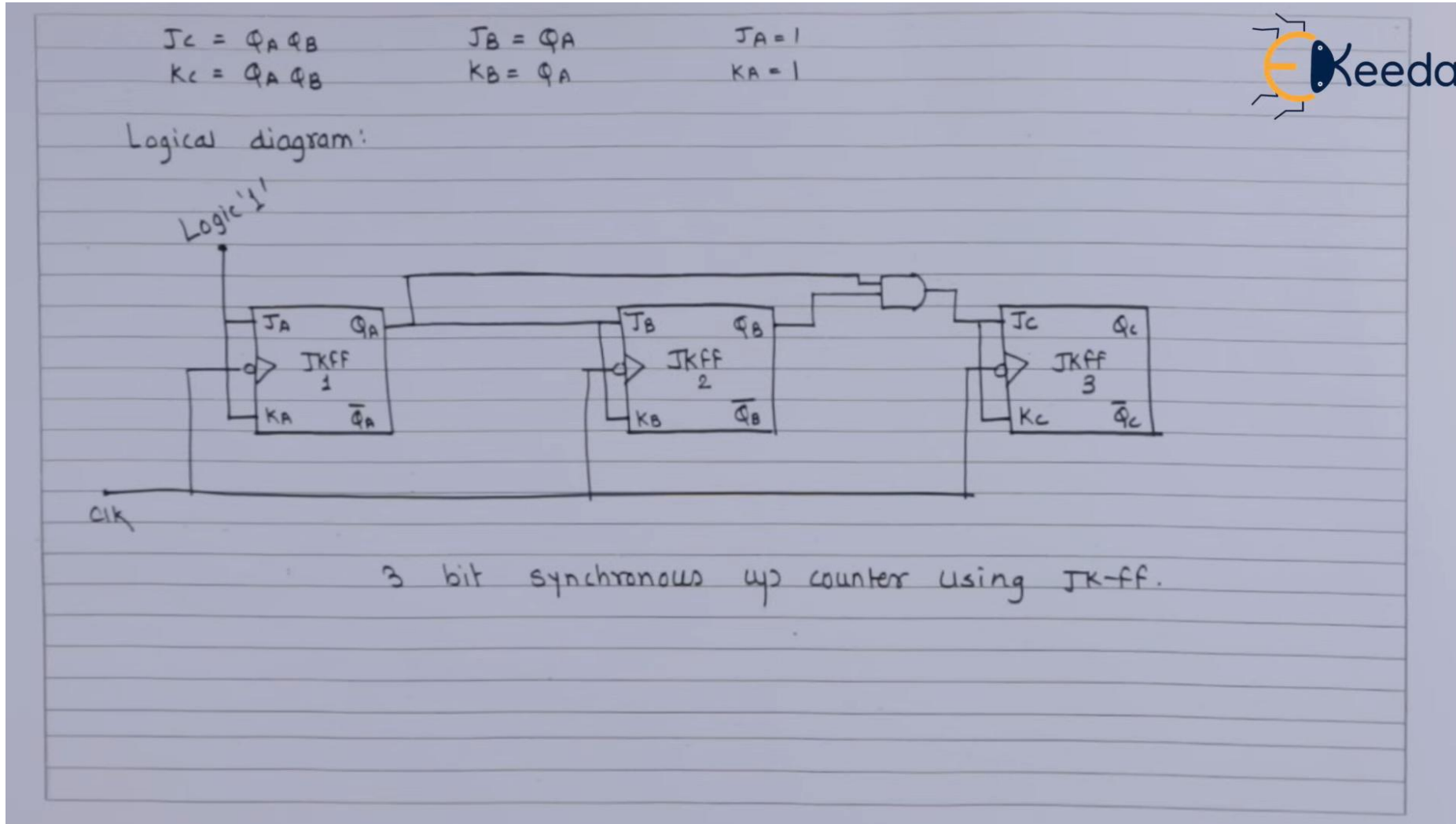
$$K_A = 1$$

K-map for K_B

$Q_C Q_B$	00	01	11	10
Q_A	X	1	1	X
	1	X	1	X

$$\therefore K_B = Q_A$$

Design a 3 bit synchronous up counter using JK Flip-Flop



Counters

- Design a MOD 10/BCD/ Decade ripple counter using JK Flip-Flop

Solution:

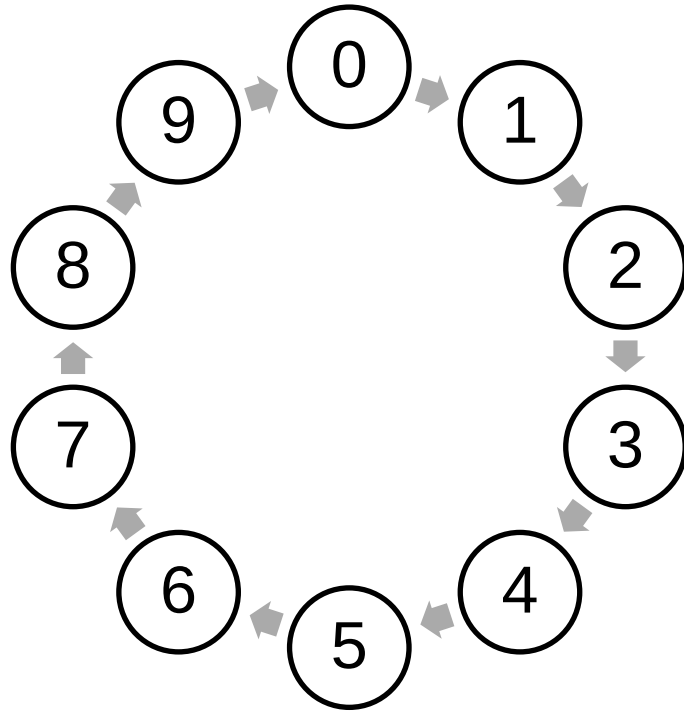
Number of States = 10

Counts from 0 to 9

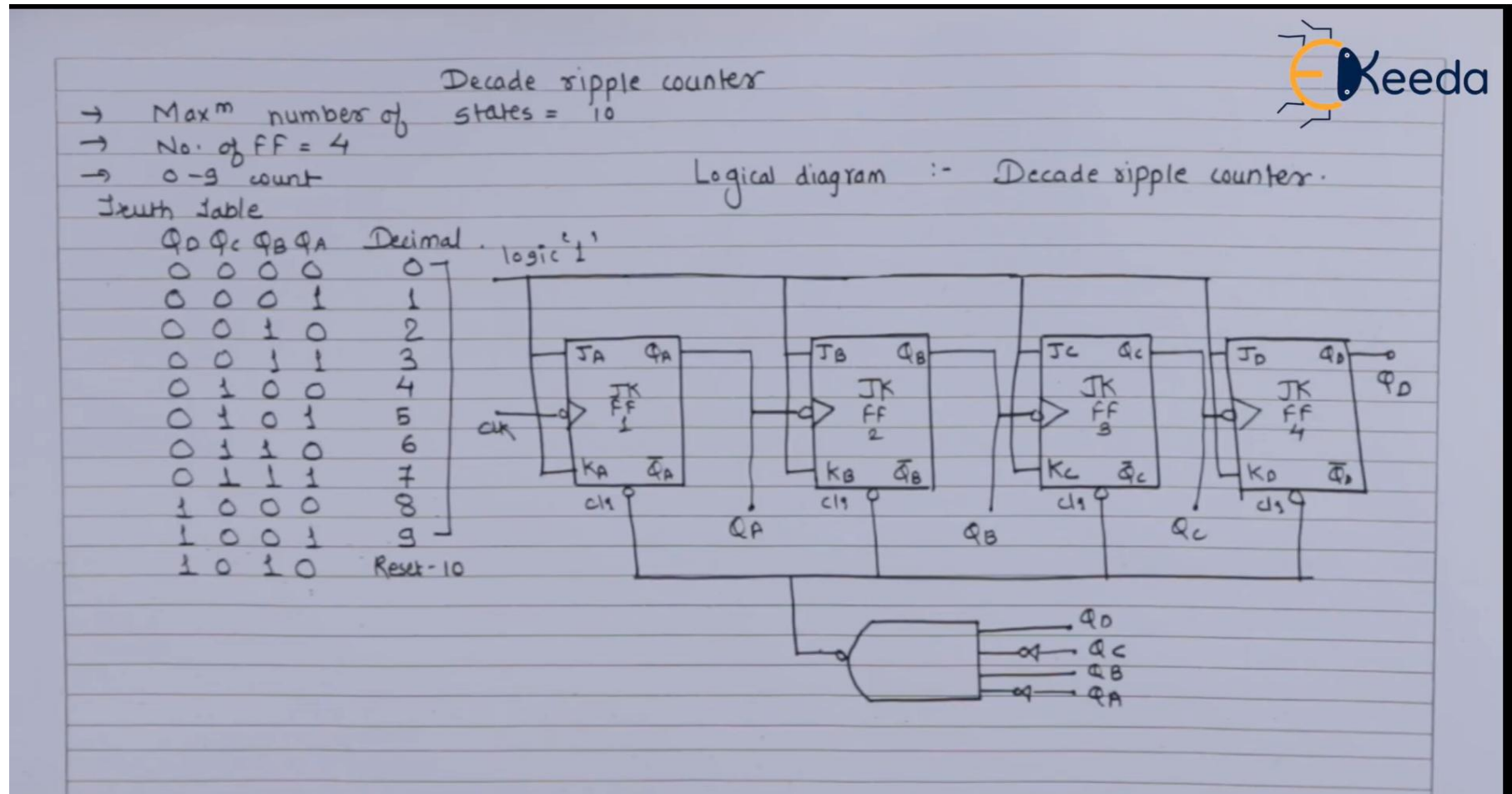
$\therefore \text{Number of Flip Flop} = N < 2^n$ where n represents the number of Flip Flop
 $10 < 2^n \therefore n = 4$

Decade counter using JK Flip Flop

State Diagram



Decade counter using JK Flip Flop



Counter with counting sequence

Design a synchronous counter with count sequence

0,3,5,6,0,...

Solution:

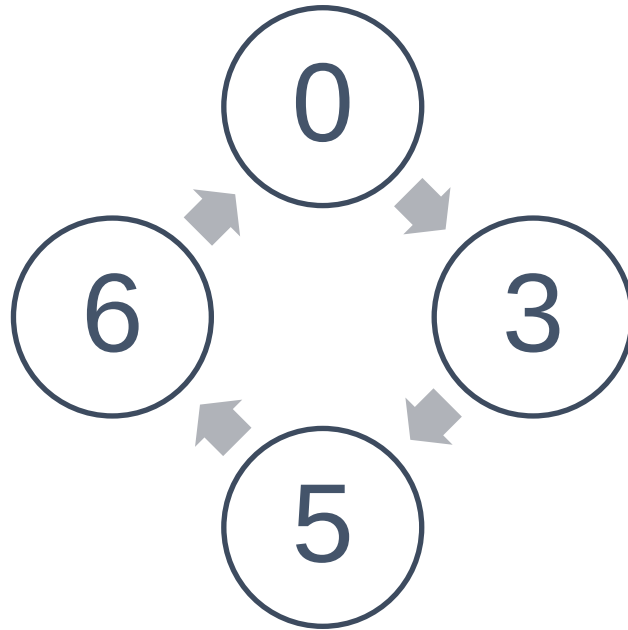
Number of States = 4

Largest count = 6

$\therefore \text{Number of Flip Flop} = N < 2^n$ where n represents the number of Flip Flop
 $6 < 2^n \therefore n = 3$

Counter with counting sequence

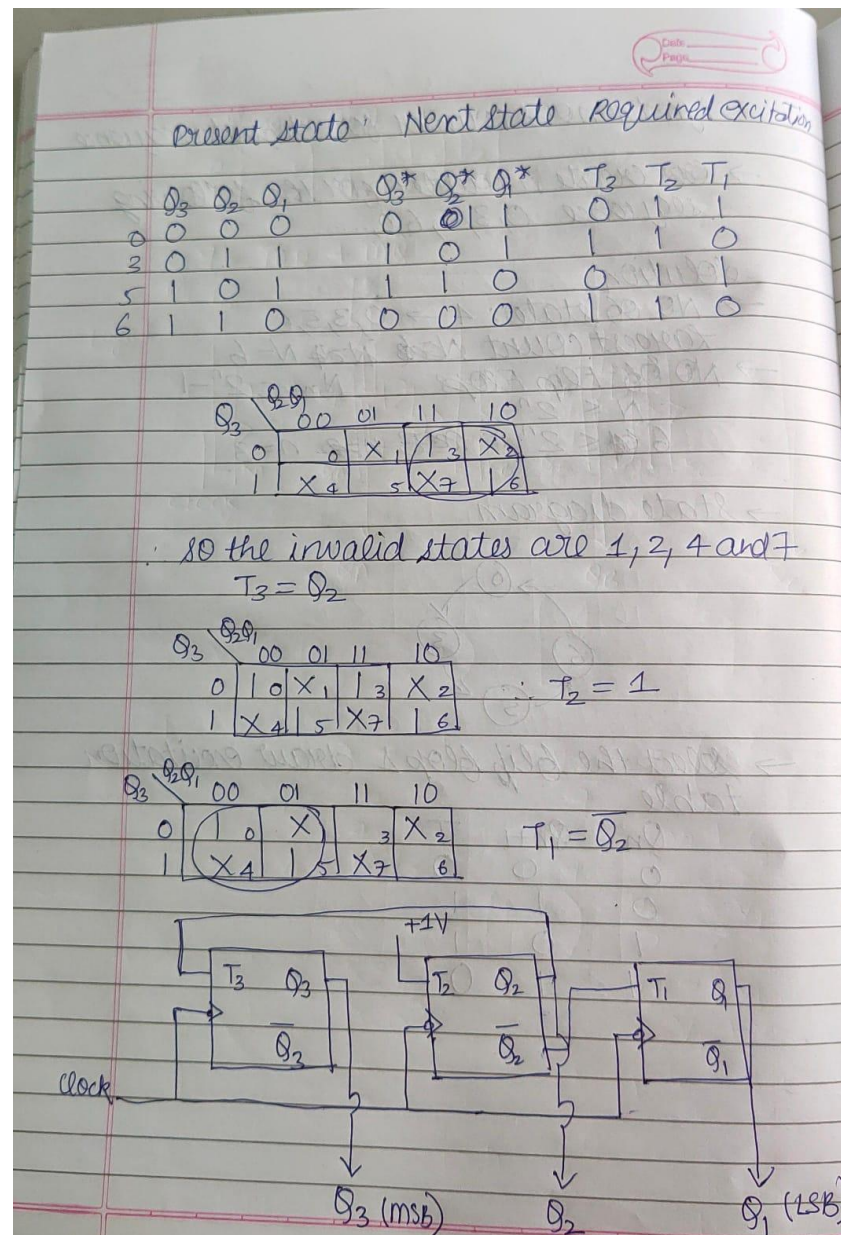
State Diagram



Q_t	$Q(t+1)$	T
0	0	0
1	0	1
0	1	1
1	1	0

Excitation Table for T Flip Flop

Counter with counting sequence



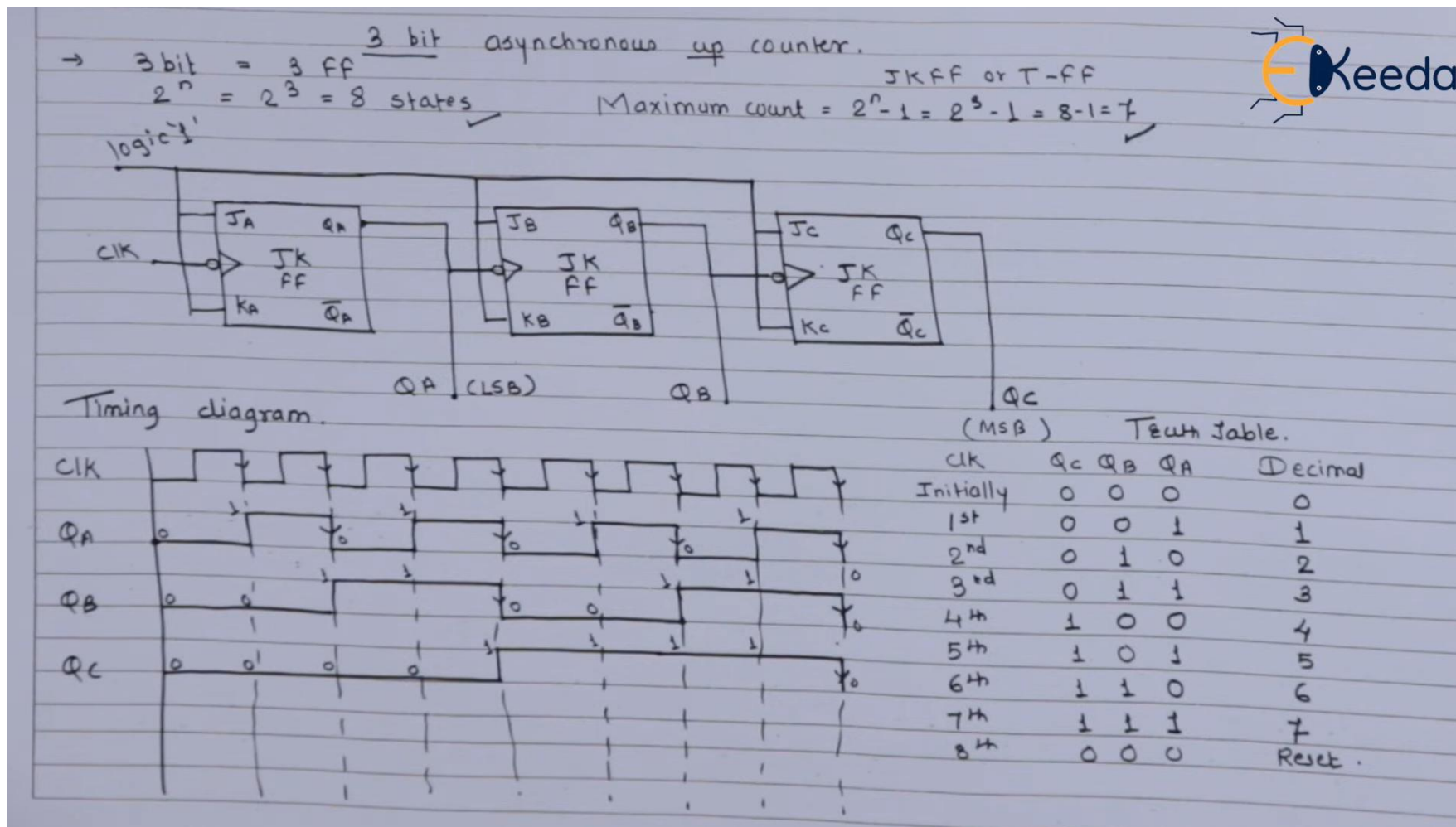
Counters

- Design a 3 bit asynchronous up counter using JK Flip-Flop

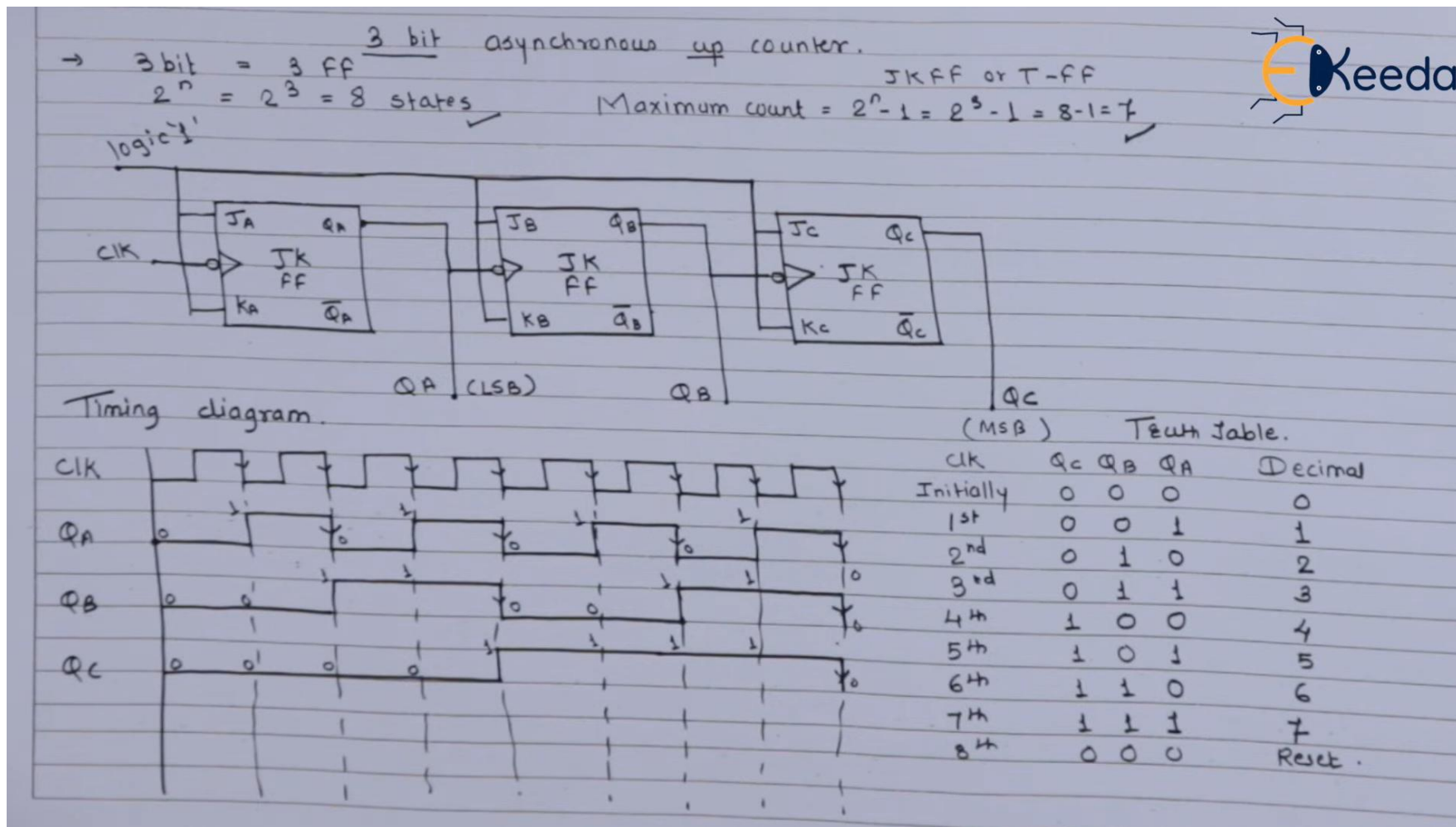
Link for reference:

<https://youtu.be/GcAOB-HyK9A?si=ye8ts0ky-aY7SLTu>

3 bit asynchronous up counter using JK Flip-Flop



3 bit asynchronous up counter using JK Flip-Flop



Counters

- Design a 3 bit asynchronous up-down counter
- Reference link: https://youtu.be/qpW8F_KBt0A?si=oB5n1dMTHA4uanL1

3 bit asynchronous up-down counter



3 bit asynchronous up down counter

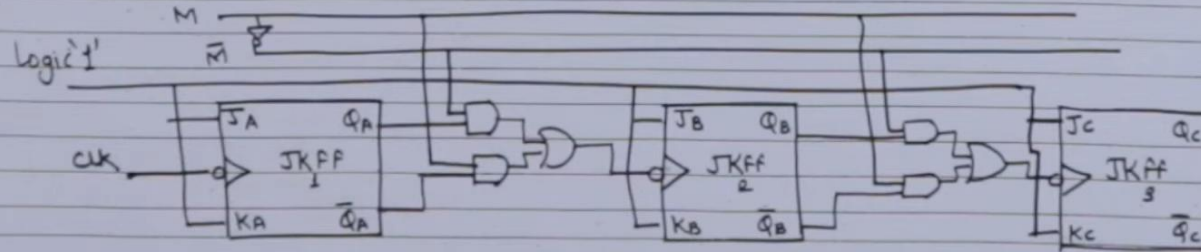
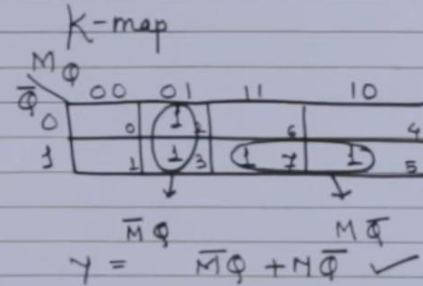
→ control input say M.

→ $M=0 \rightarrow$ up counting $\rightarrow \Phi$ connected to the clock

→ $M=1 \rightarrow$ down counting $\rightarrow \bar{\Phi}$ connected to the clock.

Truth Table

Input	output
M Φ $\bar{\Phi}$	Y
0 0 0	0
0 0 1	0
0 1 0	1
0 1 1	1
1 0 0	0
1 0 1	1
1 1 0	0
1 1 1	1



Register in Digital Electronics

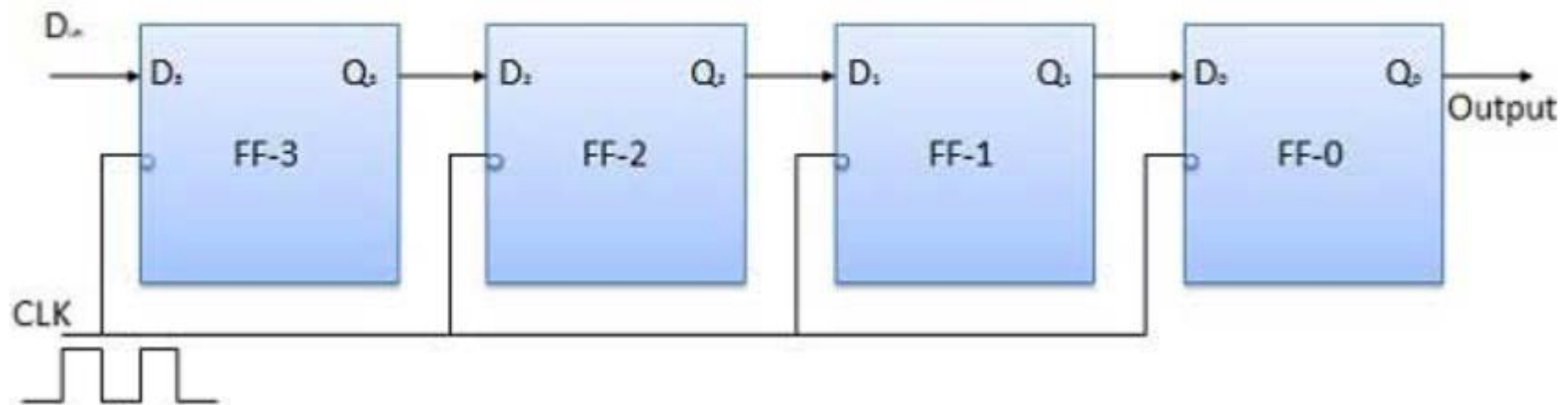
- A register is a tiny, fast storage memory within the central processing unit (CPU) or the arithmetic logic unit (ALU) of a computer.
- Registers are utilized for a variety of functions in handling and controlling instructions and data and play an important role in the operation of a computer's CPU.
- A shift register is a group of flip-flops connected in series to store and shift multiple bits of data either left or right with clock pulses.
- The registers which will shift the bits to the left are called “Shift left registers”.
- The registers which will shift the bits to the right are called “Shift right registers”.

Serial-in-Serial-Out (SISO) register

- A Serial-In Serial-Out (SISO) shift register accepts data one bit at a time through a single input line and outputs data serially, using a series of flip-flops all triggered by the same clock. The main use of a SISO is to act as a delay element.
- Data enters and leaves one bit at a time in a serial sequence.
- Typically made of multiple D flip-flops connected in series and synchronised by a common clock.

Serial-in-Serial-Out (SISO) register

Block Diagram



Serial-in-Serial-Out (SISO) register

- Let all the flip-flop be initially in the reset condition i.e. $Q_3 = Q_2 = Q_1 = Q_0 = 0$.
- If an entry of a four bit binary number 1 1 1 1 is made into the register, this number should be applied to Din bit with the LSB bit applied first.
- The D input of FF-3 i.e. D3 is connected to serial data input Din. Output of FF-3 i.e. Q3 is connected to the input of the next flip-flop i.e. D2 and so on.

Serial-in-Serial-Out (SISO) register

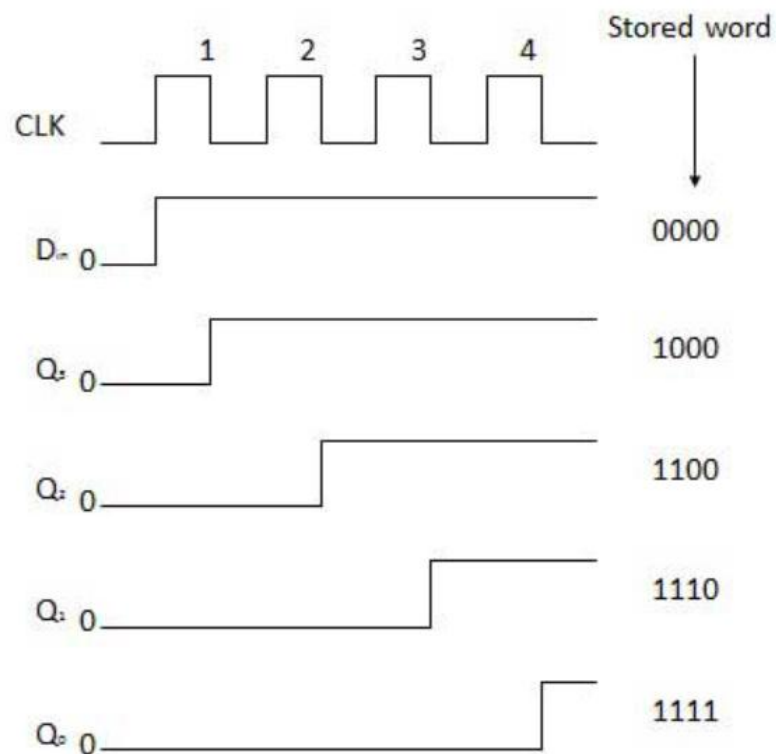
Truth Table

	CLK	$D_{in} = Q_0$	$Q_1 = D_0$	$Q_2 = D_1$	$Q_3 = D_2$	Q_4
Initially			0	0	0	0
(i)	↓	1 →	1	0	0	0
(ii)	↓	1 →	1	1	0	0
(iii)	↓	1 →	1	1	1	0
(iv)	↓	1 →	1	1	1	1

→ Direction of data travel

Serial-in-Serial-Out (SISO) register

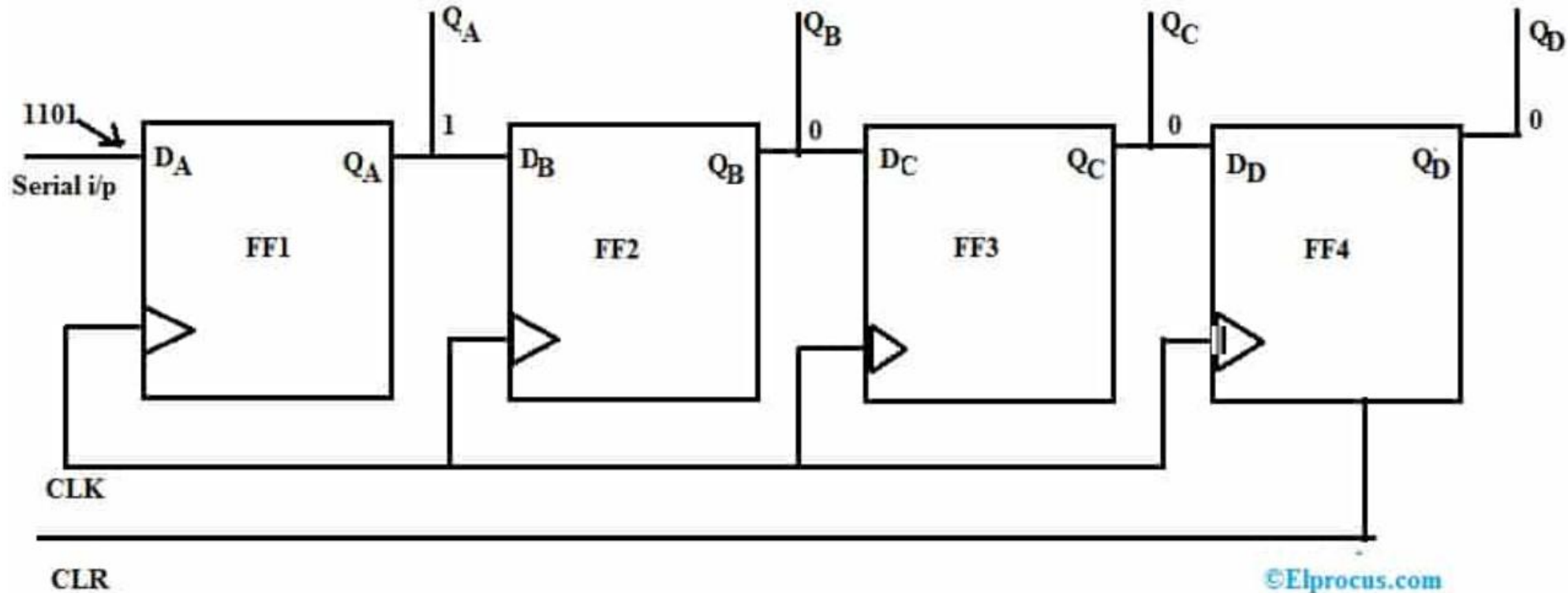
Waveforms



Serial-in-Parallel-Out (SIPO)

- In such types of operations, the data is entered serially and taken out in parallel fashion.
- Data is loaded bit by bit. The outputs are disabled as long as the data is loading.
- As soon as the data loading gets completed, all the flip-flops contain their required data, the outputs are enabled so that all the loaded data is made available over all the output lines at the same time.
- 4 clock cycles are required to load a four bit word. Hence the speed of operation of SIPO mode is same as that of SISO mode.

Serial-in-Parallel-Out (SIPO)



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SIPO Shift Register Diagram

Serial-in-Parallel-Out (SIPO)

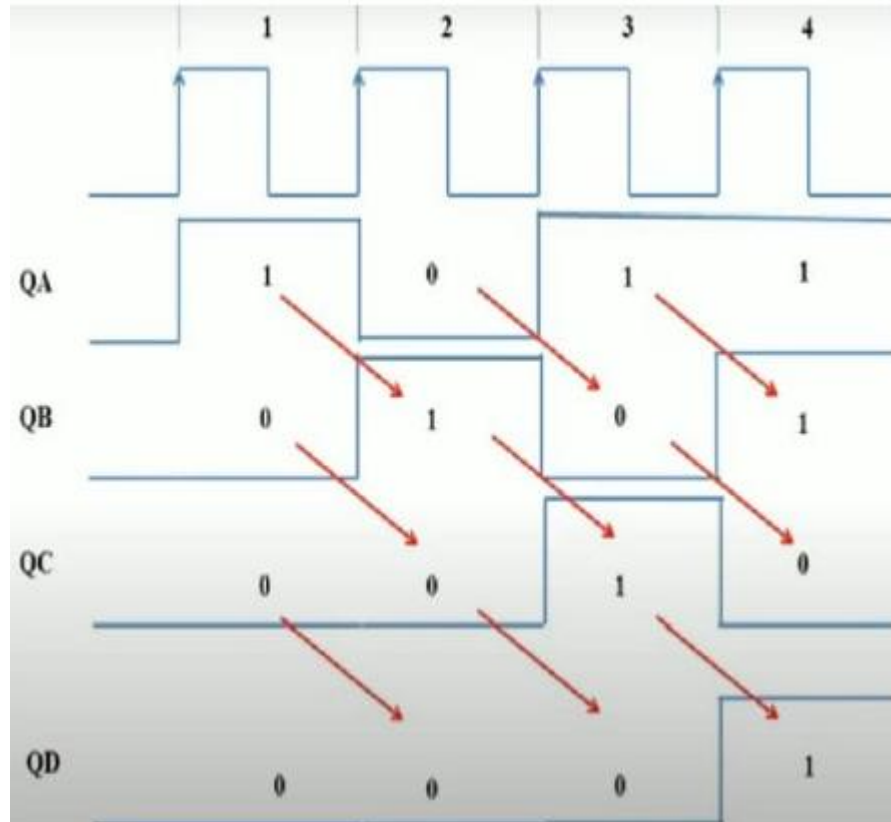
- Let us apply data '1011' to the register

CLK Pulse	QA	QB	QC	QD
0	0	0	0	0
1	1	0	0	0
2	0	1	0	0
3	1	0	1	0
4	1	1	0	1

SIPO Shift Register Truth Table

Serial-in-Parallel-Out (SIPO)

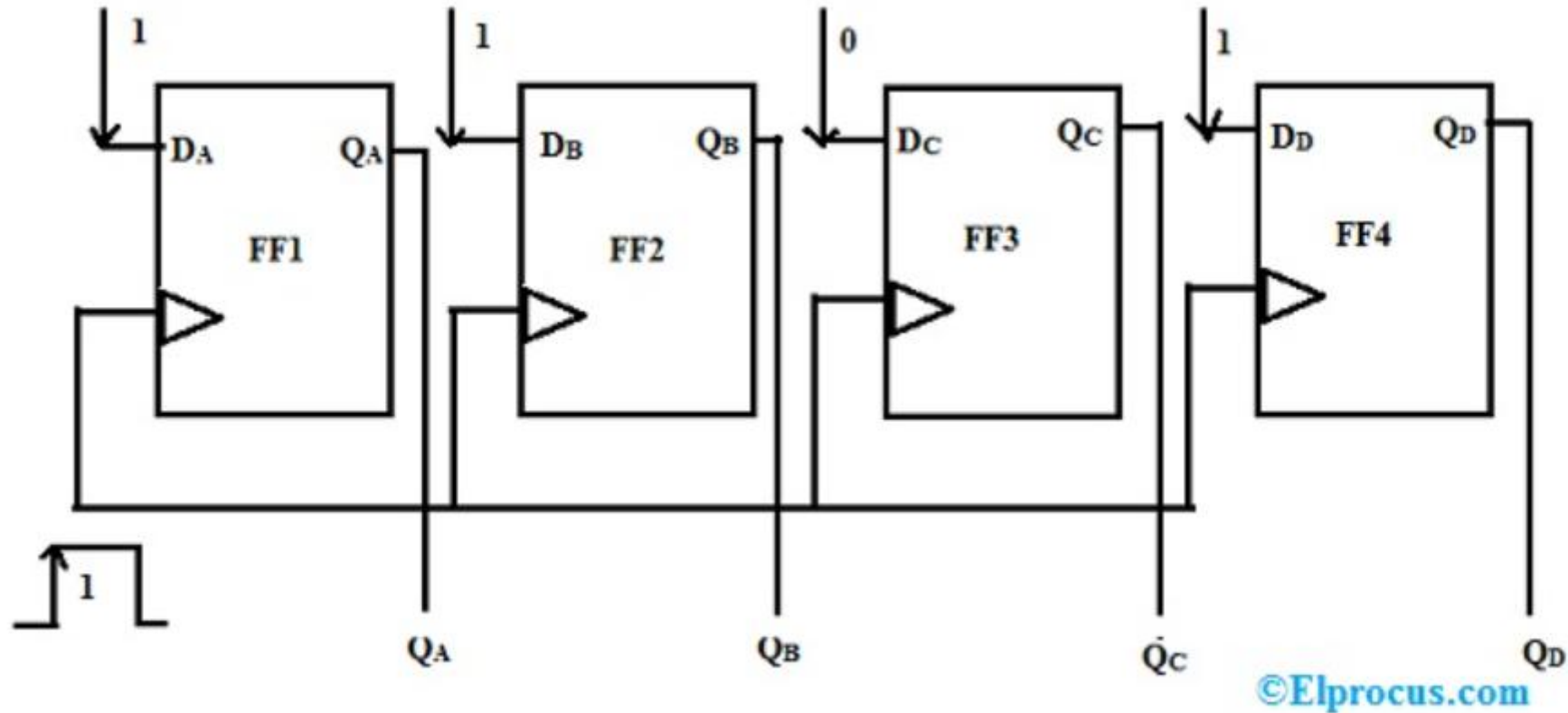
- Let us apply data '1011' to the register



Parallel-in-Parallel-Out (PIPO)

- The circuit diagram of the PIPO shift register is shown in the next slide.
- The input allowed by this type of shift register is parallel & gives a parallel output.
- This logic circuit is designed with 4 D-FFs which is shown in the diagram. In this circuit, both the CLR and CLK signals are connected to 4 D FFs.

Parallel-in-Parallel-Out (PIPO)



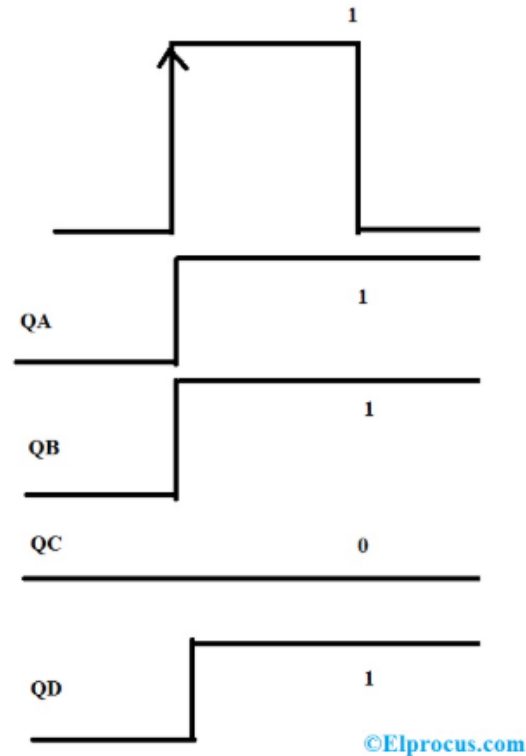
PIPO Shift Register Circuit Diagram

Parallel-in-Parallel-Out (PIPO)

- Now if we take the data input is 1101 which is loaded in each flip flop, initially, the output will become 0000.
- If we apply the first CLK pulse '1' then the input data will be shifted from input DA to QA, so the output will become 1101.
- The truth table for SIPO is shown below:

CLK Pulse	QA	QB	QC	QD
0	0	0	0	0
1	1	1	0	1

Parallel-in-Parallel-Out (PIPO)

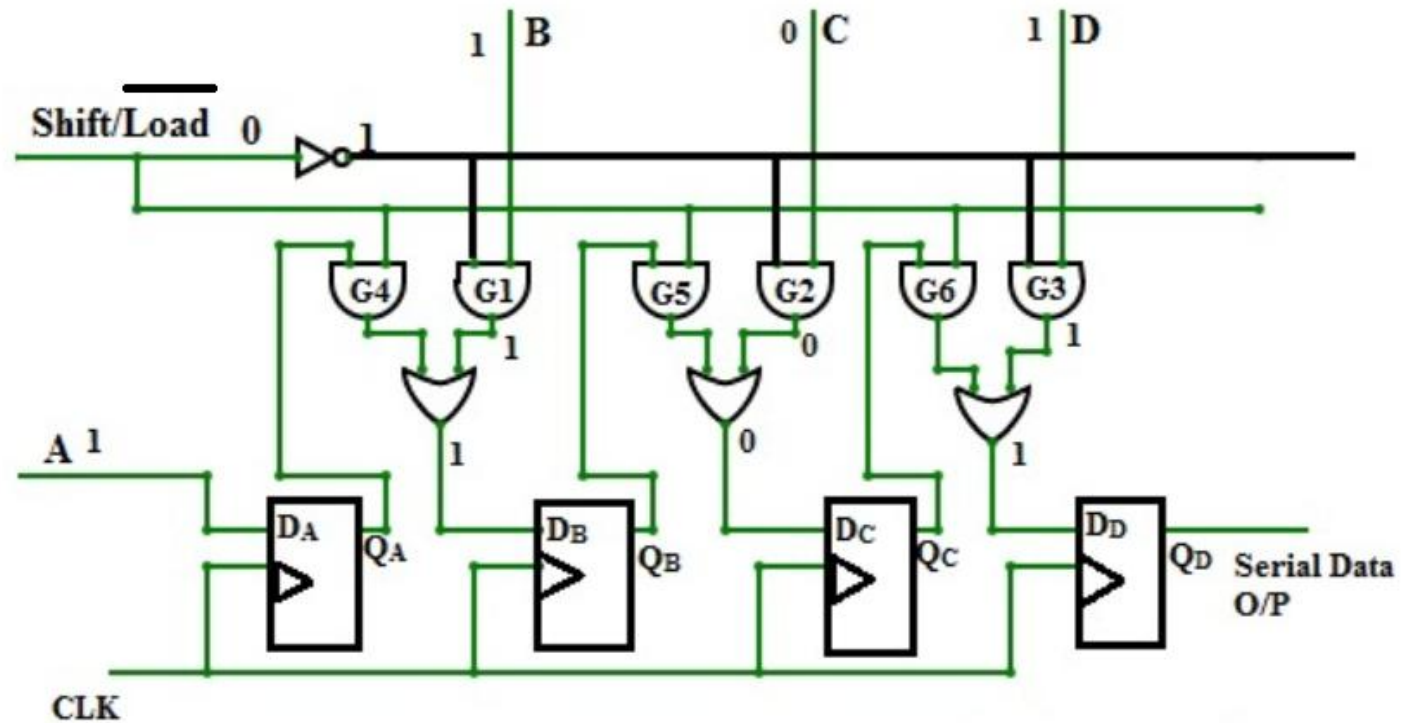


Timing Diagram

Parallel-in-Serial-Out

- The shift register which uses parallel input and generates serial output is known as the parallel input serial output shift register or PISO shift register.
- This shift register works in a reverse way to the SIPO shift register.
- In this type of shift register, the input data enters a parallel way and comes out serially. So the i/p of the second FF is the o/p of the first flip flop.
- This circuit mainly includes 4 D FFs which are connected as per the diagram shown.
- The CLK i/p signal is connected directly to all the FFs however the i/p data is individually connected to every flip flop.

Parallel-in-Serial-Out



PISO Shift Register Circuit

Parallel-in-Serial-Out

- Now the control signal applied is '0' then the data to be loaded and the data will become 1101.

D _A	D _B	D _C	D _D
1	1	0	1

- Now the control signal applied is '1' and the CLK pulse '1' is applied then the data is shifted like QA becomes '1', QB becomes '1', QC becomes '0' and QD becomes '1' as shown in the table.

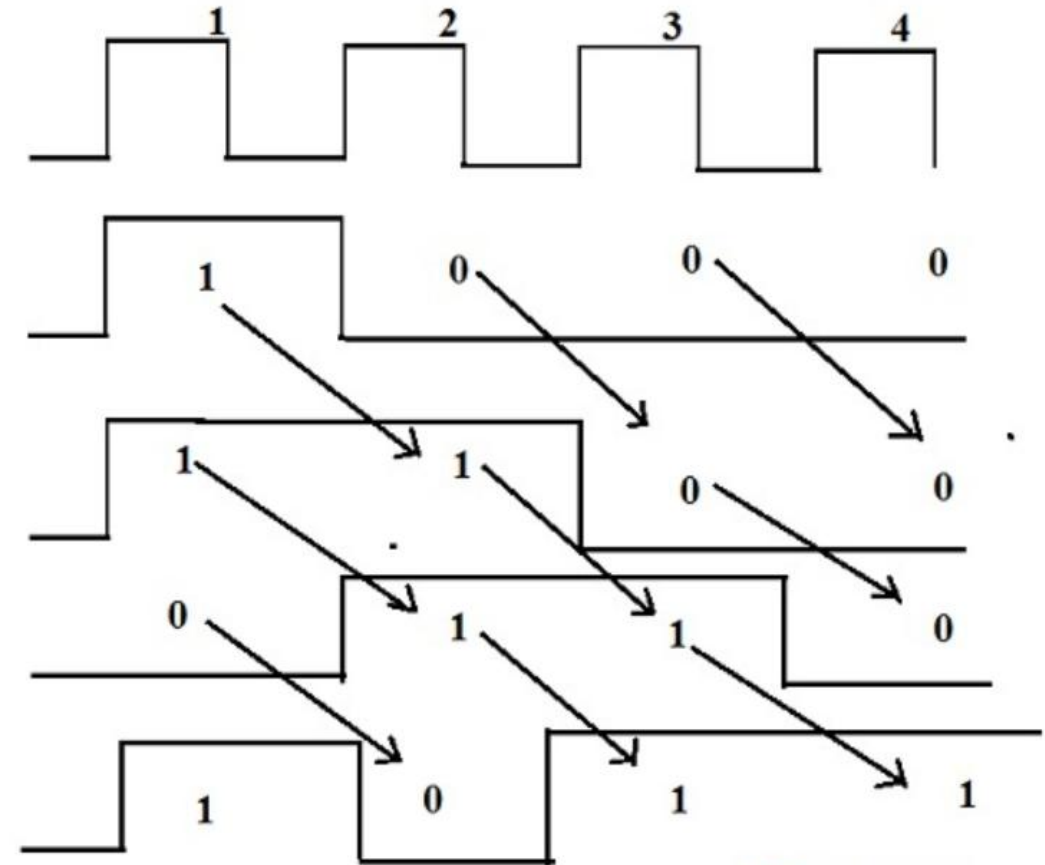
Parallel-in-Serial-Out

- Now the control signal applied is '1' and the CLK pulse '1' is applied then the data is shifted like Q_A becomes '1', Q_B becomes '1', Q_C becomes '0' and Q_D becomes '1' as shown in the table.

CLK Pulse	Q_A	Q_B	Q_C	Q_D (Data Output)
0	0	0	0	0
1	1	1	0	1
2	0	1	1	0
3	0	0	1	1
4	0	0	0	1

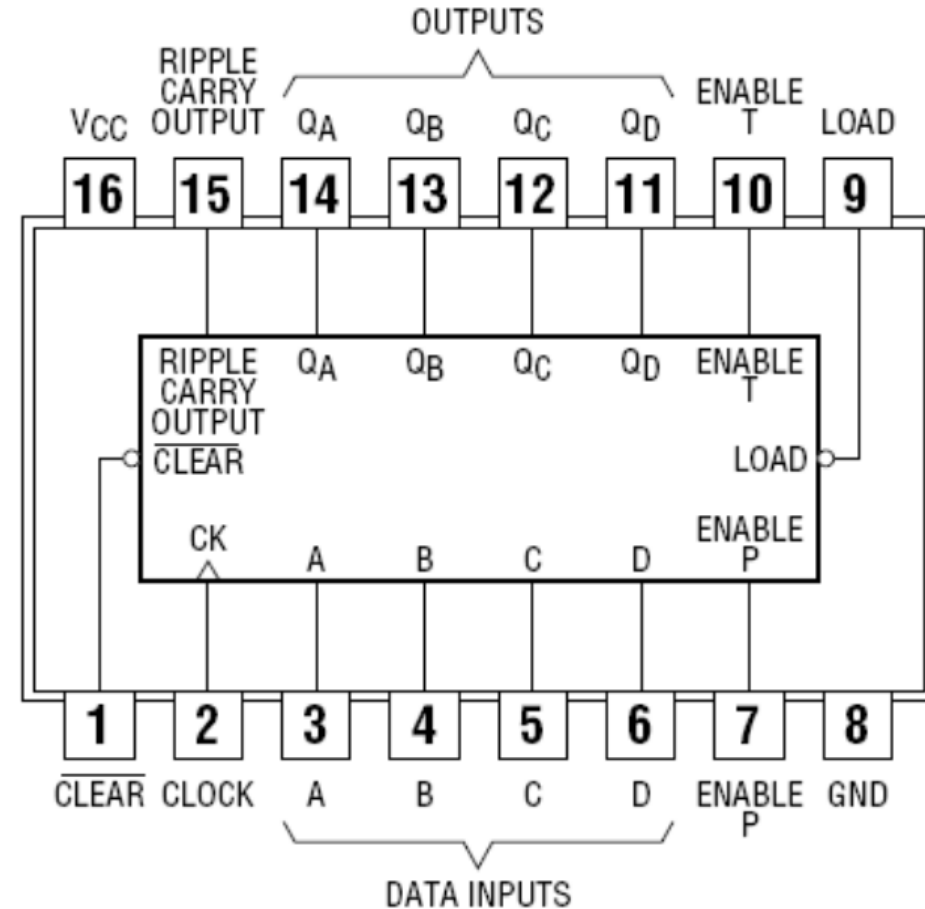
Parallel-in-Serial-Out

- The timing diagram of the PISO shift register is shown below.
- Here we are using a positive edge CLK i/p signal.
- Once the CLK pulse is applied then all the data is to be shifted from QA, QB, QC & QD.
- After the second CLK pulse, the '1' is moving to the third and fourth CLK pulses which are shown in the diagram.
- So the final data output after the fourth CLK will be 1011.



MSI SHIFT REGISTER –IC 74161

- The IC 74161 is a 16-pin IC and it is a 4 bit-binary counter IC.
- The IC has two enable pins 'ENABLE P' and 'ENABLE T' setting either of the pin to low will stop the counting asynchronously.
- RIPPLE CARRY OUTPUT is normally low and becomes high only when the counter reaches its highest value 15 in this case.
- Setting 'ENABLE T' to zero also inhibits RIPPLE CARRY OUTPUT to go from '0' to '1'



MSI SHIFT REGISTER –IC 74161

Example 8.13

Design a normal mod-12 counter using 74161.

Solution

The circuit is designed for the normal UP counting (last row of Fig. 8.28b). The Q_D and Q_C outputs through a NAND gate are connected to the Cr terminal which clears the counter as soon as the output is 1100. The states of the counter are from 0000 through 1011. The mod-12 counter is shown in Fig. 8.29.

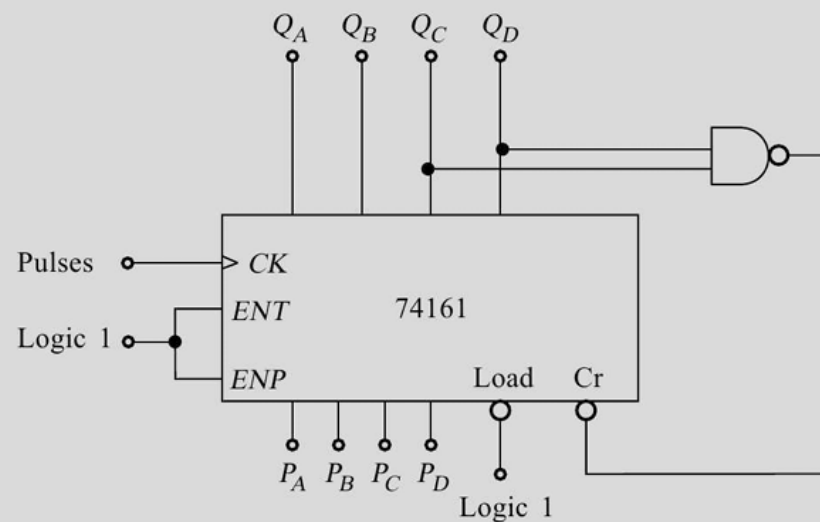


Fig. 8.29

Figure for Ex. 8.13